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Ludlow et al.

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(54) **WIRELESS COMMUNICATION LINK DEBUGGING**

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Related U.S. Application Data

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(51) **Int. Cl.**

H04W 72/04	(2009.01)
H04W 24/08	(2009.01)
H04L 12/24	(2006.01)
H04L 12/26	(2006.01)

(52) **U.S. Cl.**

CPC ***H04W 24/08*** (2013.01); ***H04L 41/5025*** (2013.01); ***H04L 43/0811*** (2013.01); *H04L 43/16* (2013.01)

(58) **Field of Classification Search**

CPC H04L 41/00; H04W 24/08; Y02B 60/50

See application file for complete search history.

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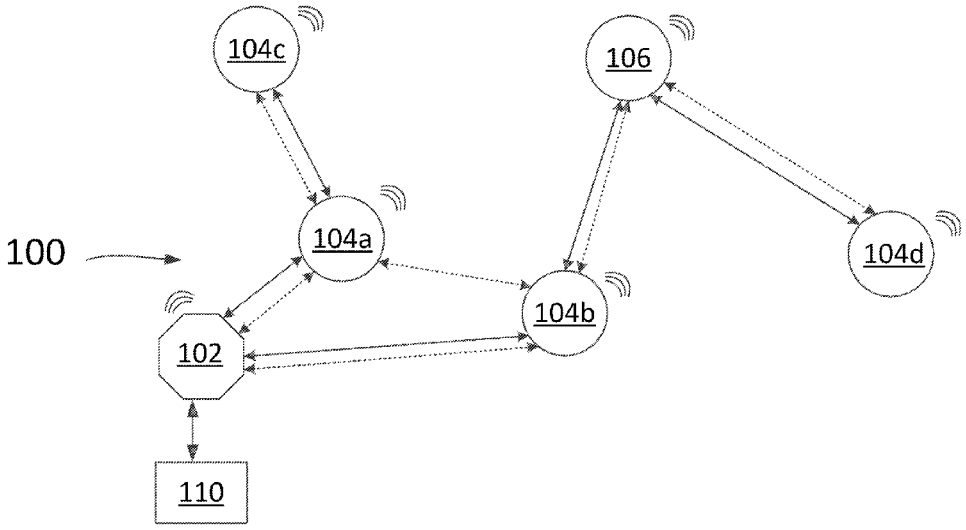
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(57) **ABSTRACT**

Provided herein is a system and method for determining the status of one or more wireless communication links in a network such as a personal area network. One or more nodes within the PAN substantially continuously monitor a plurality of quality of service (QoS) communication indicators and generate a first application link quality indicator (ALQI) based on a first subset of the plurality of QoS communication indicators according to a first time period. The node can then modify the first ALQI with a second subset of the plurality of QoS communication indicators only when a first predetermined threshold number of occurrences of the second subset of the plurality of QoS communication indicators has occurred without regard to the first time period to form a first modified ALQI. The node can then report the modified ALQI to a coordinator node that can then act upon the modified ALQI to possibly correct one or more communication link problems to which the reported ALQI pertains to.

14 Claims, 8 Drawing Sheets



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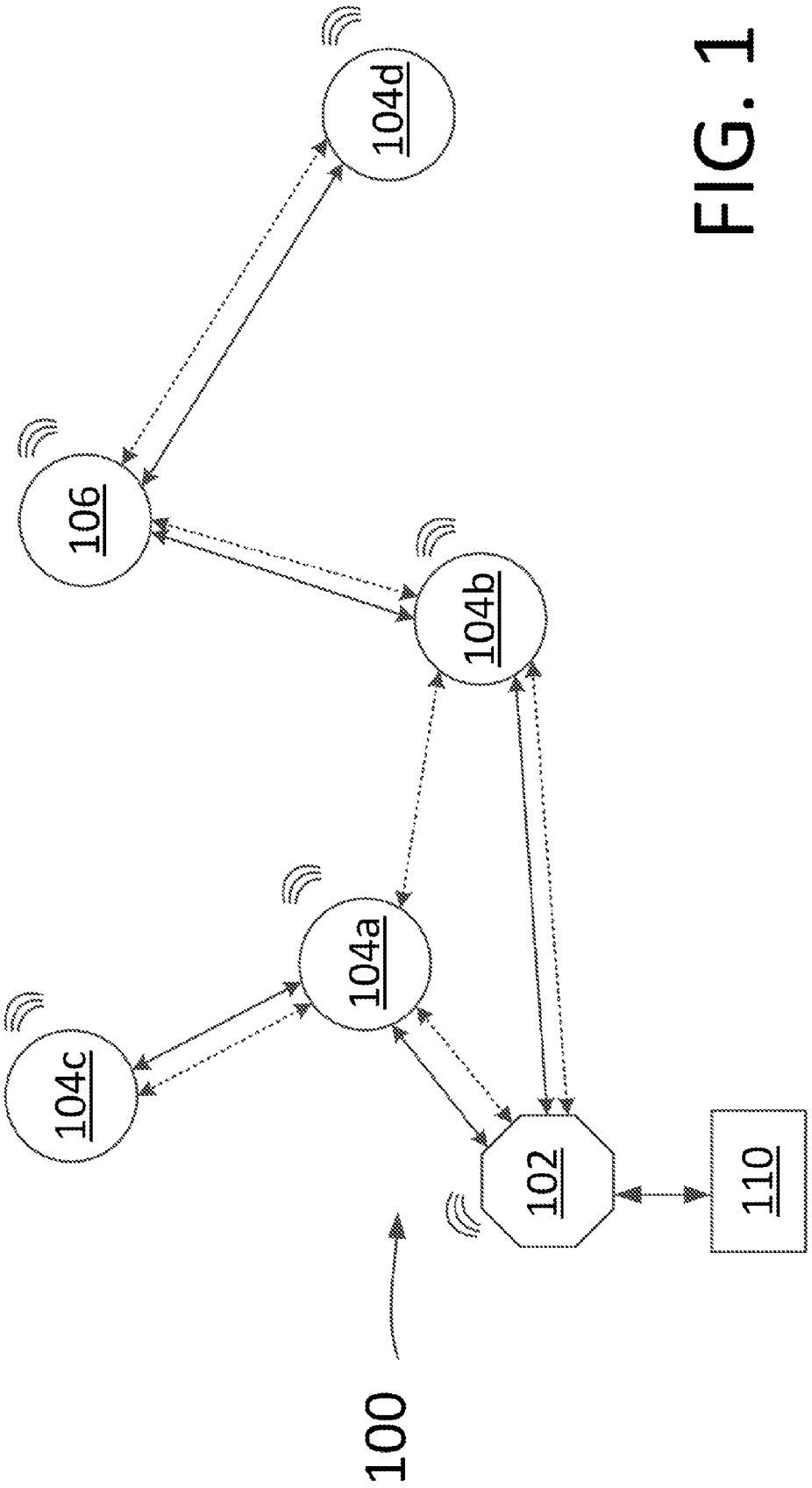


FIG. 1

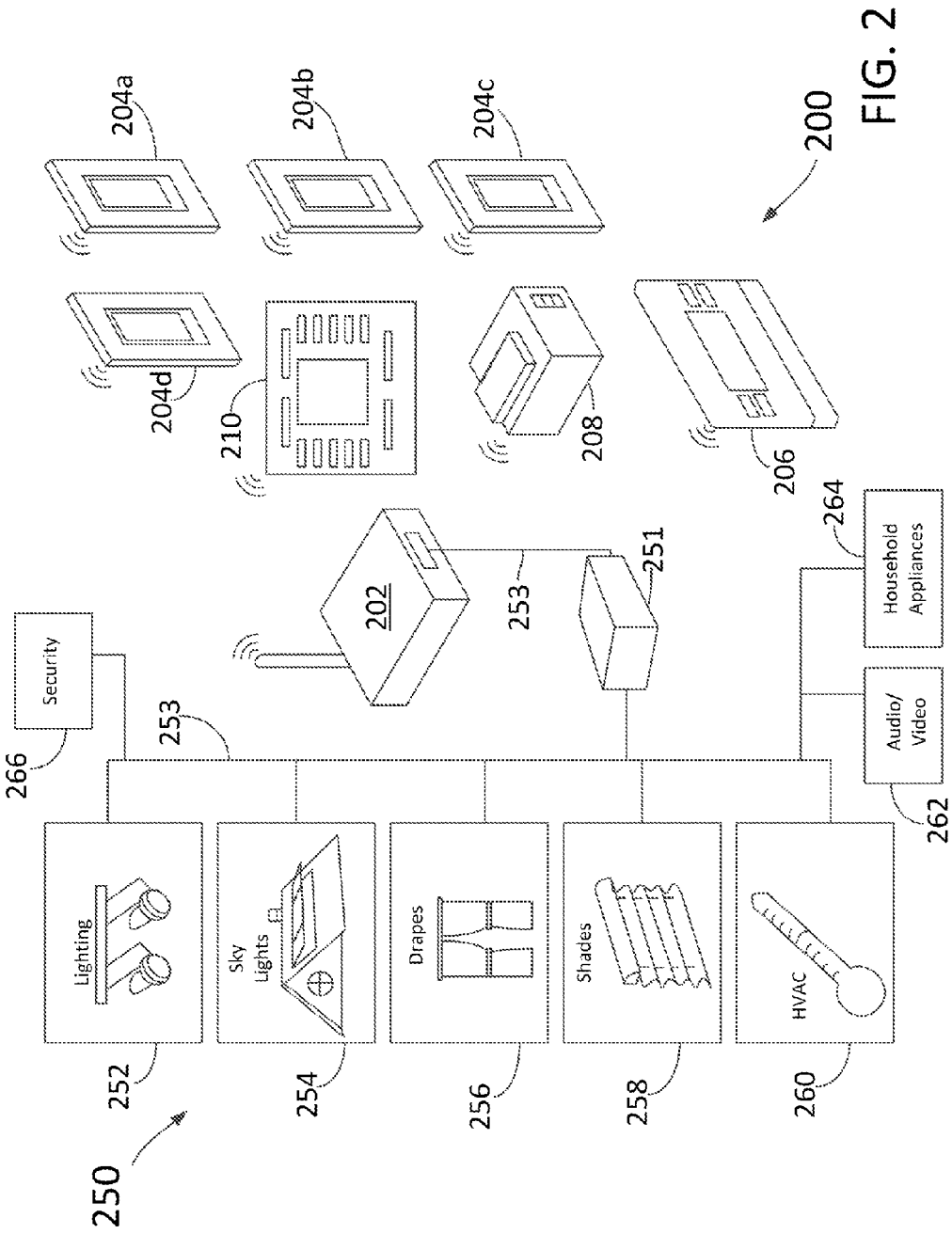
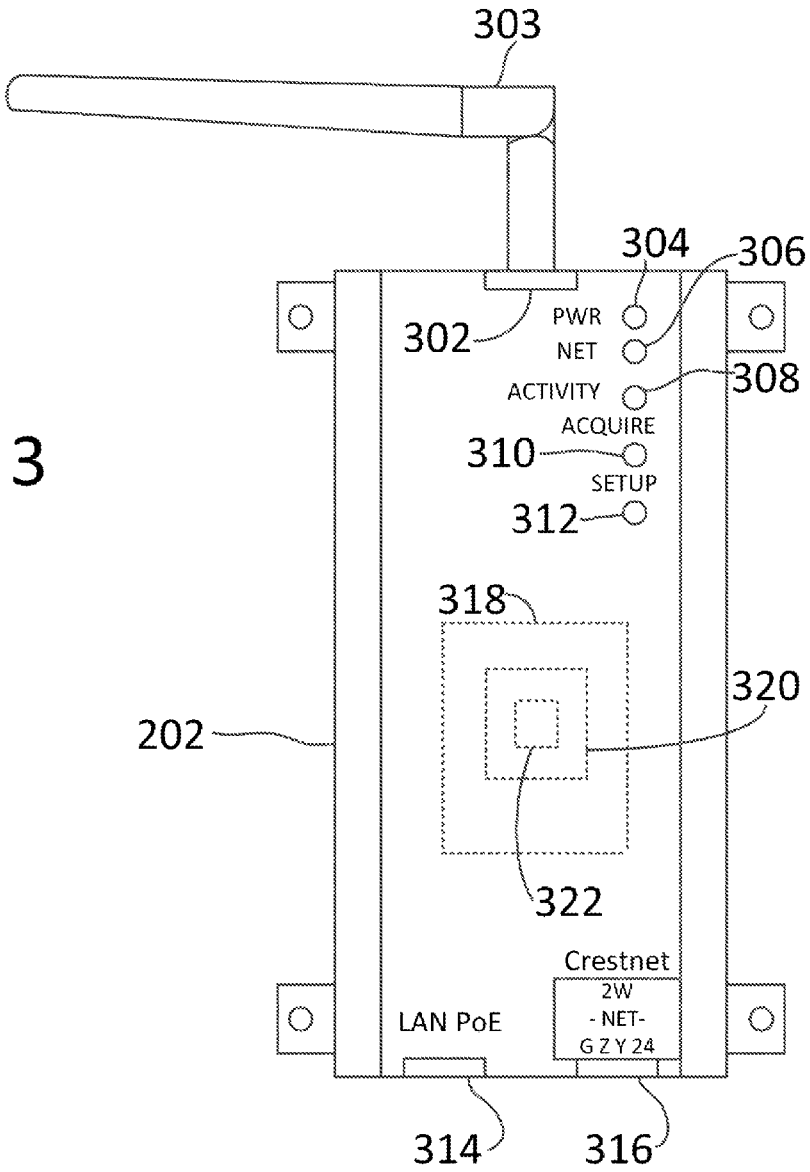

200
204a
204b
204c
204d
210
208
206
202
253
251
264
Household Appliances
Audio/ Video
262
Security
266
253
Lighting
Sky Lights
Drapes
Shades
HVAC
252
254
256
258
260
250

FIG. 2

FIG. 3


303
304
306
302
PWR
NET
ACTIVITY
308
ACQUIRE
310
SETUP
312
318
320
202
322
Crestnet
2W
- NET-
G Z Y 24
LAN PoE
314
316

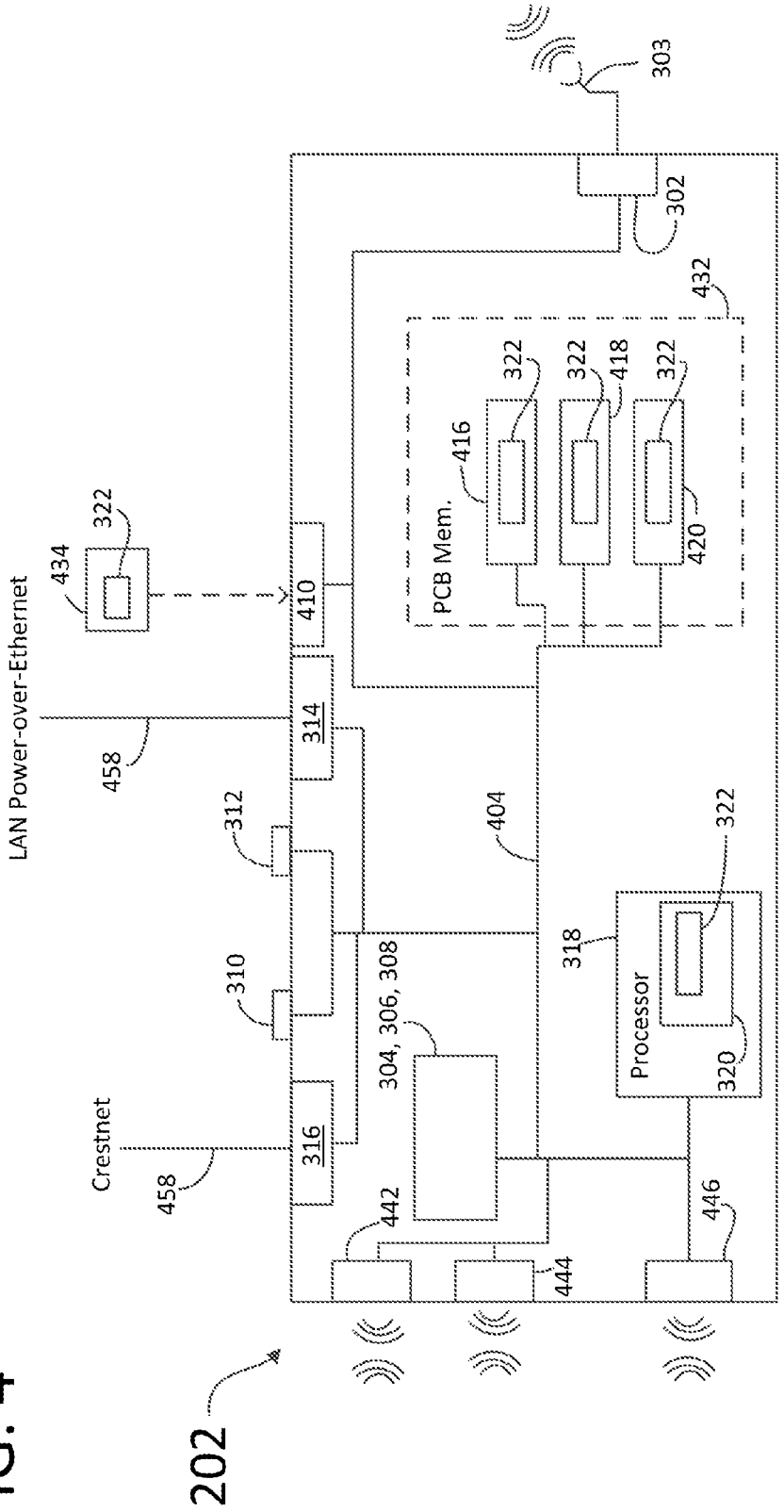

303
302
432
322
322
418
322
416
PCB Mem.
420
434
322
410
LAN Power-over-Ethernet
458
314
312
404
322
318
Processor
320
310
304, 306, 308
Crestnet
458
316
442
444
446
202

FIG. 4

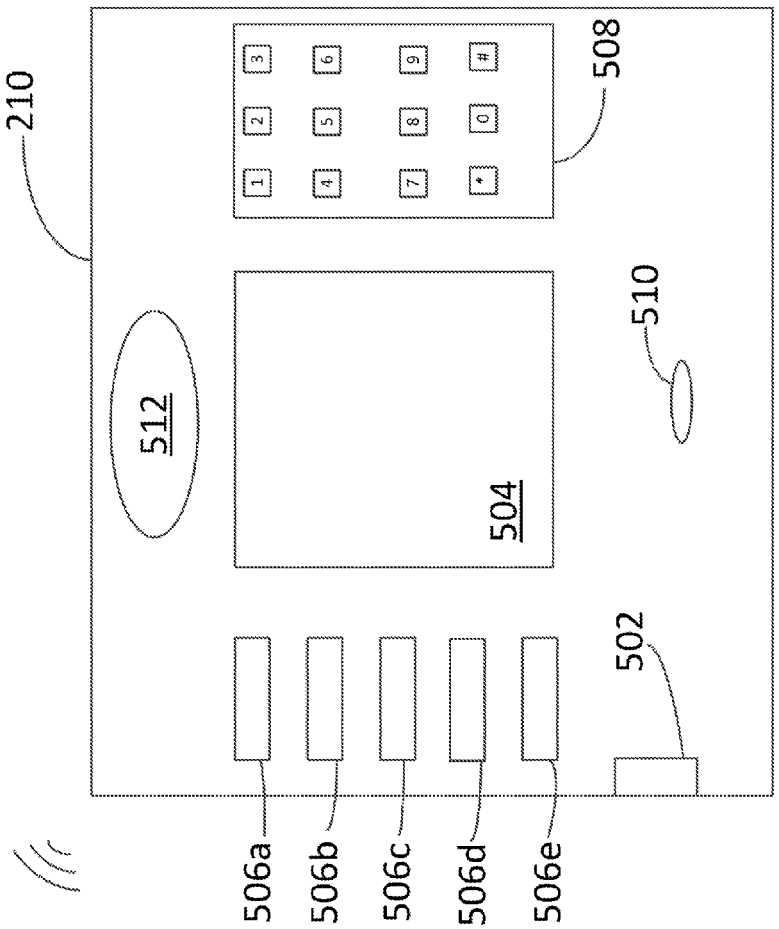


FIG. 5

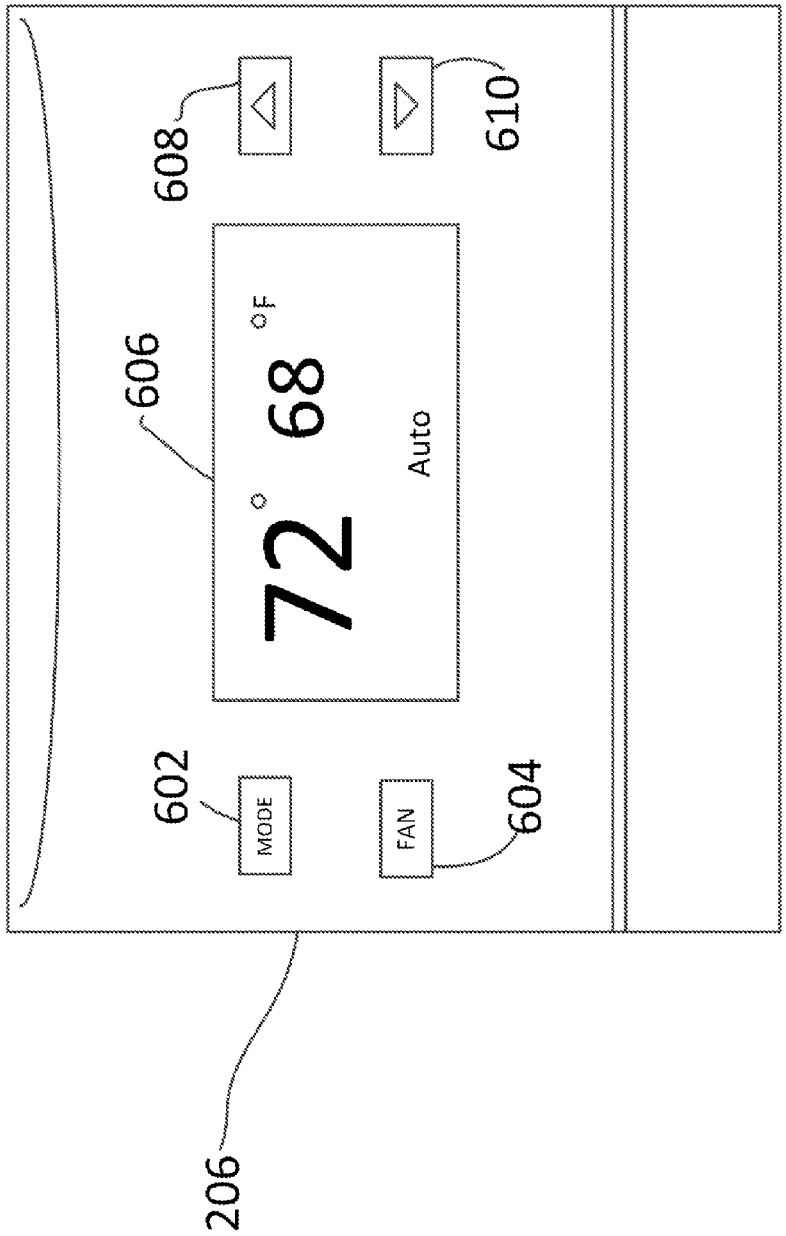


FIG. 6

		Value Range:	0-7	0-255	-100 -0	
1	2	3	4	5	6	7
ID	NODE	ALQI	ZLQI	COST	TBMT	RSSI
12	D47F	0	0	0	0	0
13	447E	0	0	0	0	0
20	6C80	255	1	1	147	-14
21	A957	0	1	1	0	0
60	F53C	0	0	0	0	0
61	10A7	0	0	0	0	0

FIG. 7A

700a

702

Value Range:		0-100%	0-100%	0-100%	0-100%	0-100%	0-100%	0-100%	0-100%
1	2	8	9	10	11	12	13	14	15
ID	NODE	MACFR	MACRR	APSFR	APSRR	RXCORR	BUSY	NTRNVR	BFEXH
12	D47F	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
13	447E	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
20	6C80	2.33%	0.00%	0.00%	0.00%	0.00%	29.99%	10.57%	0.00%
21	A957	0.00%	78.98%	0.05%	6.23%	0.00%	53.83%	0.01%	12.5%
60	F53C	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
61	10A7	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%

FIG. 7B

700b

702

	Value Range:	0-64	0-48	0-64	0-32	0-32	0-16	0-16	0-16
1	2	16	17	18	19	20	21	22	23
ID	NODE	APF	CCA	APR	MCF	MCR	RXC	NTT	MTT
12	D47F	0	0	0	0	0	0	0	0
13	447E	0	0	0	0	0	0	0	0
20	6C80	0	16	0	0	0	0	0	0
21	A957	0	0	22	47	0	32	15	12
60	F53C	0	0	0	0	0	0	0	0
61	10A7	0	0	0	0	0	0	0	0

FIG. 7C

700c

702

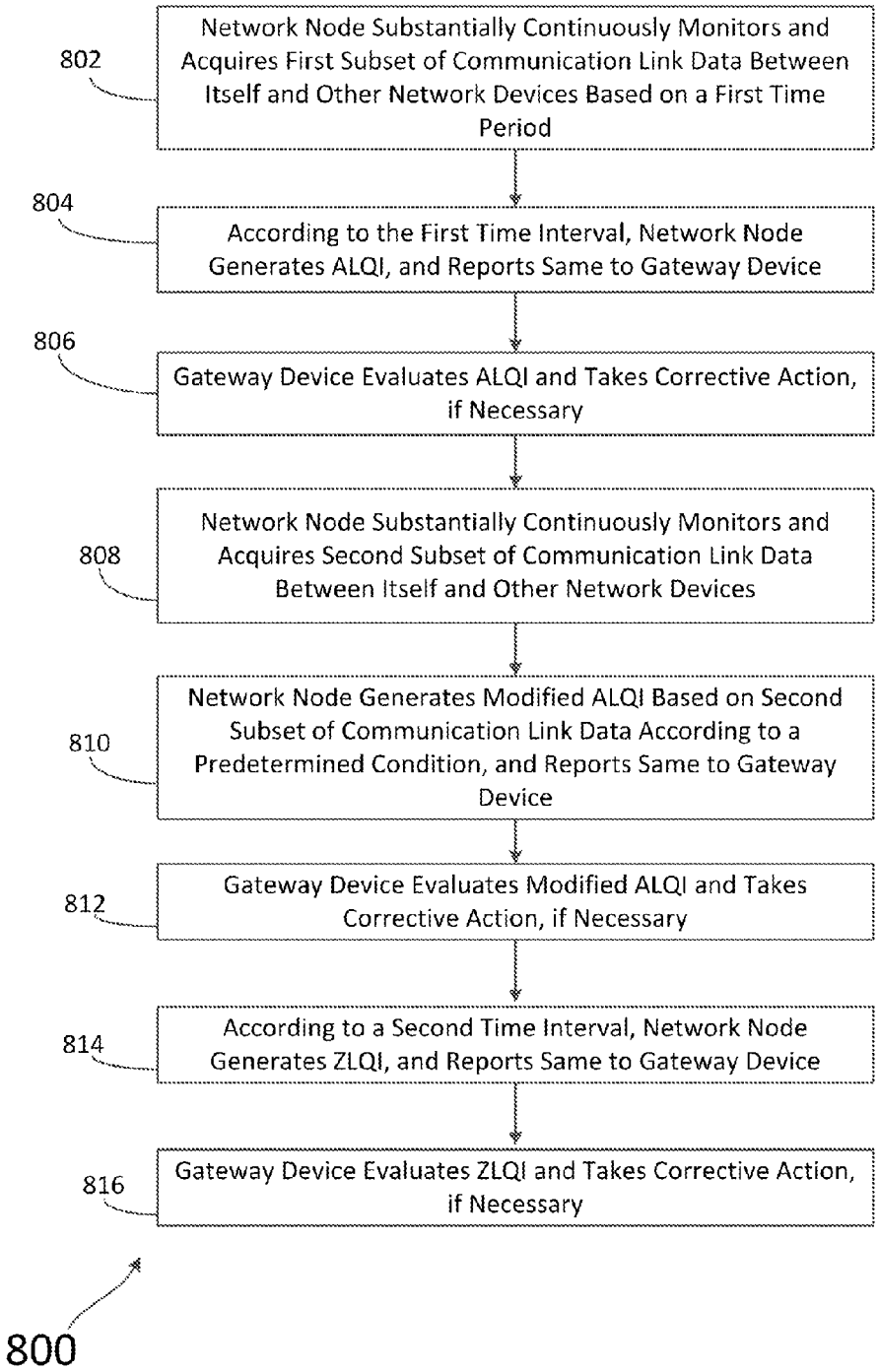

802
Network Node Substantially Continuously Monitors and Acquires First Subset of Communication Link Data Between Itself and Other Network Devices Based on a First Time Period
804
According to the First Time Interval, Network Node Generates ALQI, and Reports Same to Gateway Device
806
Gateway Device Evaluates ALQI and Takes Corrective Action, if Necessary
808
Network Node Substantially Continuously Monitors and Acquires Second Subset of Communication Link Data Between Itself and Other Network Devices
810
Network Node Generates Modified ALQI Based on Second Subset of Communication Link Data According to a Predetermined Condition, and Reports Same to Gateway Device
812
Gateway Device Evaluates Modified ALQI and Takes Corrective Action, if Necessary
814
According to a Second Time Interval, Network Node Generates ZLQI, and Reports Same to Gateway Device
816
Gateway Device Evaluates ZLQI and Takes Corrective Action, if Necessary
800

FIG. 8

WIRELESS COMMUNICATION LINK DEBUGGING

PRIORITY INFORMATION

The present application claims priority under 35 U.S.C. §119(e) to U.S. Provisional Patent Application Ser. No. 62/025,469, filed 16 Jul., 2014, and U.S. Provisional Patent Application No. 62/043,495, filed 29 Aug., 2014, the entire contents of both of which are expressly incorporated herein by reference.

BACKGROUND OF THE INVENTION

Technical Field

Aspects of the embodiments relate to wireless networks and more specifically to systems, devices and methods for determining the quality of one or more communications links in a wireless network.

Background Art

Control networks in homes are increasingly popular installations. A network of sensors and control devices distributed throughout a residential or commercial building can control and provide information for various elements of the building. For example, one or more control networks in a home can control and monitor lighting, heating, ventilation and air conditioning (HVAC), and audio-visual (AV) equipment.

Increasingly, these control networks comprise one or more personal area networks (PANs) that are organized and in communication with other PANs and the control system. Such communications can include conflict resolution mechanisms, among others. Generally, these PANs allow each device to communicate with each other, thereby providing multiple possible routes between nodes over which a packet can be routed. This is in contrast to traditional networks in which devices generally communicate through a master or a repeater.

FIG. **1** illustrates a block diagram of a first PAN **100**. PAN **100** is a network topology in that uses a source routing format. In the source routing format, only the nodes in a proscribed route are actively involved in propagating the message. In some cases, a broadcast message can be used that involves all or most of the other nodes, but that typically does not happen. Nonetheless, all of the nodes can cooperate in the distribution of data in PAN **100**, but typically do not, as described above. PAN **100** typically has a self-healing capability that enables data rerouting when one node breaks down or a connection goes bad. As a result, Pan **100** is typically quite "robust" or reliable, as there is often more than one path between a source and a destination in PAN **100**. Although mostly used in wireless situations (shown as dashed lines), PAN **100** is also applicable to wired networks (solid lines) and software interaction. PANs **100** are applicable to data only, as power cannot be effectively be transferred wirelessly.

While PANs **100** can be implemented with either a flooding technique or a routing technique to transmit messages, the former is less preferred. That is because the flooding technique incurs massive bandwidth overhead in exchange for their lightweight protocol overhead, making them very inefficient. When using a routing technique, the message is propagated along a path, by hopping from one PAN **100** node **104** to the next node **104** until the destination is reached (e.g., first PAN **100** node **104***a* to second PAN **100** node **104***b*, and so on). To ensure the availability of all of its paths, a routing network must allow for continuous connections and reconfiguration around broken or blocked paths using self-healing algorithms.

When nodes **102**, **104**, and **106** are all connected to each other in PAN **100**, it is commonly referred to as a fully connected network. Node **102** is a special node, and can be referred to as coordinator node **102**; nodes **104** can be further defined as fully functioning device (FFD) nodes (FFD node **104**), or reduced function device (RFD) nodes (RFD node **106**). Node **110**, which is not always required, can be referred to as diagnostic node **110**. Diagnostic node **110** provides a user with the ability to ascertain the communication link status within PAN **100**. Diagnostic node **110** is substantially similar in function as diagnostic console **210** shown and described in reference to FIG. **2**, and thus a detailed discussion is made in reference thereto. Coordinator node **102**, FFD node **104**, and RFD node **106** are described in greater detail below. A fully connected network can be costly, as either a wired connection is required between each node (or computer) or a wireless interface needs to be installed. Of course, the wireless interface can save wiring costs, but can also prove to be less reliable (and slower) under some conditions, as those of skill in the art can appreciate.

As shown in FIG. **1**, PAN **100** comprises coordinator node **102** in wired communication with FFD nodes **104***a* and **104***b* (solid lines), and in wireless communication (dashed lines) FFD nodes **104***a* and **104***b*. FFD node **104***a* is in wired communications with FFD node **104***c*, and is in wireless communication with FFD nodes **104***b* and **104***c*. FFD node **104***b* is in wired and wireless communications with RFD node **106**, and RFD node **106** is in both wired and wireless communications with FFD node **104***d*. While all of the nodes of PAN **100** have wireless communications capabilities, for certain reasons (distance, interference, among other reasons), not all nodes can communicate wirelessly with all other nodes. It is noted, however, that RFD node **106**, while in both wired and wireless communications with FFD node **104***d*, cannot route messages to FFD **104**, although it can communicate its own messages to FFD **104**.

Communication protocols govern the operation of wireless PANs **100** by governing network formation, communication, interferences and other operational characteristics. For example, one such protocol is the Zigbee protocol from the Zigbee Alliance. The Institute of Electric and Electronic Engineers (IEEE) 802.15.4 standard specifies the physical layer and media access control for low-rate wireless personal area networks (LR-WPANs). It is maintained by the IEEE 802.15 working group, which defined it in 2003. It is the basis for the ZigBee, International Society of Automation (ISA) **100.11***a*, WirelessHART, and MicroChip's "MiWi" specifications, each of which further extends the standard by developing the upper layers which are not defined in IEEE 802.15.4. Another wireless communication protocol is the infiNET EX protocol from Crestron Electronics, Inc. of Rockleigh, N.J. Based on ZigBee PRO technology, infiNET EX is an extremely reliable and affordable protocol that is employed by 2-way wireless light dimmers, light switches, thermostats, keypads and other wireless sensors and control devices, described below in regard to FIG. **2**.

Wireless mesh network devices within in a Zigbee or infiNET network (from hereon in referred to as PAN **100**) can be classified as coordinator node **104**, FFD node **104**, and RFD node **106**. PAN Coordinator node (coordinator node) **102** is the device that is responsible for starting the formation of PAN **100**. There is only one coordinator **102** in

any PAN **100**. Coordinator node **102** chooses the PAN identification (PAN-ID), the 16-bit value that is used to uniquely define PAN **100**.

FFD node **104** is a device that can route communication packets as part of its normal operation. FFD node **104** can also be called a "router." FFD nodes **104** form the network topology by establishing communications links to other devices (both FFD nodes **104** and RFD nodes **106**). FFD nodes **104** can act as a sensor, actuator or other type of node.

RFD node **106** is a device that cannot route communication packets. They are always associated with a single FFD node **104**, and can only participate in PAN **100** as an end node, or leaf node.

Wireless networks, such as Zigbee or infiNET networks, are formed by an initialization process. Typically, many of configuration properties can be preconfigured prior to initialization. By preconfiguring these properties, an administrator can determine the size and shape of the network. For example, the maximum number of links from PAN coordinator **102** to the furthest device can influence the shape and size of the network. The maximum number of devices that can connect to FFD node **104** can also determine the shape and size of the network. According to an aspect of the embodiments, coordinator node **102** can specify the number of devices that can link to FFD node **104**, as well as how many of these devices can be routers themselves. According to further aspects of the embodiments, however, device applications can decide what node type they are, and they can define how many RFD devices can route through them.

Coordinator **102** is responsible for initializing wireless PAN **100**. First coordinator **102** chooses a suitable radio channel on which to operate. Next, coordinator **102** assigns a PAN ID to the network. The PAN ID can be pre-configured or alternatively can be chosen by detecting other networks operating in the same frequency channel and choosing a PAN ID that does not conflict with theirs. Coordinator **102** also assigns a network address to itself. Typically, this address is zero (0).

Once coordinator **102** is finished configuring itself, it starts itself in a PAN coordinator mode. It is then prepared to field queries from other devices wishing to join the network through a commissioning process. Other devices, both FFD nodes **104** and RFD nodes **106**, can join the network once PAN **100** has been created by coordinator **102**. Coordinator **102** has the capability to allow other nodes to join. Additionally, FFD nodes **104** can also allow other nodes to join.

Devices join PAN **100** through a commissioning process. First, the new device (i.e., the new node) scans available channels to find a wireless network. The device then selects a discovered network by choosing a router or network coordinator on that network. Once this is done, the device sends a join message to the desired router or network coordinator, either FFD node **104** on PAN **100** or coordinator node **102**.

However, PANs **100** do not always work as well upon commissioning or changes to the operating environment. Due to the complexity of the system and the transparent nature of their operation, it is not always easy to diagnose issues within the system.

Typically, the information available via communications from stack software is not useful for troubleshooting problems encountered in the field. This is largely due to the typical performance metrics being reported from the stack software being based on a very narrow spectrum of data that is itself frequently not useful. Such stack software analyzes communication events it is involved in, such as message delivery, message delivery failure, receive energy levels, and received data errors. As those of skill in the art can appreciate, the "stack" implements the lowers layers of communication, and provides an interface that can access a limited set of information it maintains, such as the ZLDU value, and event counters that can be used in system-level debugging. One example of such a stack software analysis program has been developed by EmbernetZNet; it includes a pre-compiled library, as well as some source files, that can be readily implemented to perform stack layer analysis.

Installers and field support individuals are left with a "pass/fail" type result with no indication of the cause of problems. Further, in certain applications, even a "pass/fail" test is not effective.

Accordingly, there is a need for an improved system of diagnosing issues within a personal area network such as mesh network **100**.

SUMMARY OF THE INVENTION

It is to be understood that both the general and detailed descriptions that follow are exemplary and explanatory only and are not restrictive of the invention.

DISCLOSURE OF INVENTION

Accordingly, a need exists for an improved system, device and method for diagnosing communications issues in a wireless personal area network such as a mesh network. Aspects of the embodiments provide these advantages and others not specifically mentioned above but described in the sections to follow.

An object of the embodiments is to substantially solve at least the problems and/or disadvantages discussed above, and to provide at least one or more of the advantages described below.

It is therefore a general aspect of the embodiments to provide systems, methods, and modes for debugging a wireless network that will obviate or minimize problems of the type previously described.

This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

Further features and advantages of the aspects of the embodiments, as well as the structure and operation of the various embodiments, are described in detail below with reference to the accompanying drawings. It is noted that the aspects of the embodiments are not limited to the specific embodiments described herein. Such embodiments are presented herein for illustrative purposes only. Additional embodiments will be apparent to persons skilled in the relevant art(s) based on the teachings contained herein.

According to a first aspect of the embodiments, a method for determining an operational status of a wireless network system is provided comprising monitoring a plurality of quality of service (QoS) communication indicators by a node of at least two or more nodes of the network system, generating a first application link quality indicator (ALQI) by the node based on a first subset of the plurality of QoS communication indicators according to a first time period, and modifying the first ALQI with a second subset of the plurality of the plurality of QoS communication indicators only when a first predetermined threshold number of occurrences of the second subset of the plurality of QoS commu-

nication indicators has occurred without regard to the first time period to form a first modified ALQI. According to the first aspect of the embodiments, the method further comprises transmitting the modified ALQI by the first node to a second node in the network system, and wherein the first node is any one of a plurality of network nodes that can include a fully functioning node, and a reduced functioning node, and further wherein, the second node is a gateway node, and further wherein the method further comprises, determining a corrective action by the gateway device based on either or both of the ALQI and modified ALQI.

According to a second aspect of the embodiments, a method for determining an operational status of a wireless network system is provided, comprising monitoring a plurality of quality of service (QoS) communication indicators by a first node of at least two or more nodes of the network system, generating a first application link quality indicator (ALQI) by the first node based on a first subset of the plurality of QoS communication indicators according to a first time period, and modifying the first ALQI with a second subset of the plurality of the plurality of QoS communication indicators only when a first predetermined threshold number of occurrences of the second subset of the plurality of QoS communication indicators has occurred without regard to the first time period to form a first modified ALQI.

According to the second aspect of the embodiments, the method further comprises transmitting the modified ALQI by the first node to a second node in the network system, and wherein the first node is any one of a plurality of network nodes that can include one or more fully functioning nodes, and one or more reduced functioning nodes, and further wherein, the second node is a gateway node, and further wherein the method further comprises determining a corrective action to correct communication link errors by the gateway device based on either or both of the ALQI and modified ALQI.

According to the second aspect of the embodiments, the corrective action can include one or causing the first node to change channels, increase transmission power, and increase receiver sensitivity, and the gateway node can communicate one or both of the received ALQI and modified ALQI to a diagnostic console connected via a wireless or wired connection to the gateway node.

Still further according to the second aspect of the embodiments, the step of generating the first ALQI comprises evaluating the following expression—

$$ALQI=255-\min[\Sigma ALQIP,255],$$

wherein ALQIP comprises the first subset of the plurality of QoS communication indicators.

According to the second aspect of the embodiments, the step of generating the modified ALQI comprises evaluating the following expression—

$$ALQI=255-\min[\Sigma ALQIP,255],$$

wherein ALQIP comprises the first and second subset of the plurality of QoS communication indicators.

Still further according to the second aspect of the embodiments, the number of occurrences of the second subset of the plurality of QoS communication indicators is four. According to the second aspect of the embodiments, the first subset of QoS communication indicators comprises fields **16-22** of a link debug data reporting packet that is generated and transmitted by the first node to the second node, and the second subset of QoS communication indicators comprise field **23** of the link debug data reporting packet.

According to the second aspect of the embodiments, field **16** of the link debug data reporting packet is defined as an APF field, and is based on an APS failure rate (APSFR), and is assigned a range of values from 0 to 64, and wherein APSFR is defined as the percent of application messages that fail to be delivered from the first node to the second node, field **17** of the link debug data reporting packet is defined as an CCA field, and is based on a BUSY rate, and is assigned a range of values from 0 to 48, and wherein the BUSY rate is an approximate percentage of time that airspace around the first node is considered "busy" and therefore unable to communicate, field **18** of the link debug data reporting packet is defined as an APR field, and is based on an APS layer retry rate (APSRR), and is assigned a range of values from 0-64, and wherein APSRR is the percentage of APS send events that are retries for an existing message, field **19** of the link debug data reporting packet is defined as an MCF field, and is based on a MAC failure rate (MACFR), and is assigned a range of values from 0 to 32, and wherein MACFR is the percentage of MAC transmission attempts that fail, field **20** of the link debug data reporting packet is defined as an MCR field, and is based on a MAC retry rate (MACRR), and is assigned a range of values from 0 to 32, and wherein MACRR is the percent of MAC transmission attempts that are retries, field **21** of the link debug data reporting packet is defined as an RXC field, and is based on an amount of corrupt received messages (RXCORR), and is assigned a range of values from 0 to 16, field **22** of the link debug data reporting packet is defined as an NTT field, and is based on a network turnover (NTRNVR), and is assigned a range of values from 0 to 16, and wherein NTRNVR is the amount of turnover in the first node's neighbor table, and field **23** of the link debug data reporting packet is defined as an MTT field, and is based on time between map-to-one route requests (TBMT), and is assigned a range of values from 0 to 16, and wherein the TBMT is an average number of seconds between receiving map-to-one requests received.

According to the second aspect of the embodiments, the MTT can be calculated according to the following expression—

$$MTT_{ALQIP}=(180-TBMT)\times 3/32$$

wherein, MTT is not reported unless a number of occurrences of TBMT is greater than or equal to four, and further wherein MTT is determined based on the last four occurrences of TBMT, regardless of the number of TBMT occurrences.

Still further according to the second aspect of the embodiments, the first time period is about one minute, and wherein the method further comprises generating a Zigbee link quality indicator (ZLQI) based on a third subset of the plurality of the plurality of QoS communication indicators, and wherein the third subset of QoS communication indicators comprises fields **5-15** of a link debug data reporting packet that is generated and transmitted by the first node to the second node, and wherein ZLQI is maintained over a runtime of the first node.

According to a third aspect of the embodiments, the methods as described herein can be used in a network as manufactured by Creston Electronics Inc., of Rockleigh, N.J.

According to a fourth aspect of the embodiments, a system channel can be changed in view of the ALQI and modified ALQI field data value to improve overall system communications. According to the fourth aspect of the embodiments, the system channel is changed in response to the QoS data being substantially continuously collected.

BRIEF DESCRIPTION OF DRAWINGS

The above and other objects and features of the embodiments will become apparent and more readily appreciated from the following description of the embodiments with reference to the following figures. Different aspects of the embodiments are illustrated in reference figures of the drawings. It is intended that the embodiments and figures disclosed herein are to be considered to be illustrative rather than limiting. The components in the drawings are not necessarily drawn to scale, emphasis instead being placed upon clearly illustrating the principles of the aspects of the embodiments. In the drawings, like reference numerals designate corresponding parts throughout the several views.

FIG. **1** illustrates a block diagram of a mesh network and communication paths therein.

FIG. **2** illustrates a wireless personal area network that implements a system and method for diagnosing communication links within the personal area network according to aspects of the embodiments.

FIG. **3** illustrates a wireless gateway device for use in the wireless personal area network of FIG. **2** according to aspects of the embodiments.

FIG. **4** illustrates a detailed block diagram of the main components of the wireless gateway device of FIG. **3** according to an aspect of the embodiments.

FIG. **5** illustrates a diagnostic console for use in the wireless personal network as shown in FIG. **1** according to aspects of the embodiments.

FIG. **6** illustrates a thermostat for use in the wireless personal network as shown in FIG. **1** according to aspects of the embodiments.

FIGS. **7**A, **7**B, and **7**C illustrate three tables of link debug data for several nodes of a personal area network such a Zigbee network for use in accordance with aspects of the embodiments.

FIG. **8** illustrates a flowchart of a method for diagnosing communication links in personal area network according to aspects of the embodiments.

DETAILED DESCRIPTION OF THE INVENTION

The embodiments are described more fully hereinafter with reference to the accompanying drawings, in which embodiments of the inventive concept are shown. In the drawings, the size and relative sizes of layers and regions may be exaggerated for clarity. Like numbers refer to like elements throughout. The embodiments may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the inventive concept to those skilled in the art. The scope of the embodiments is therefore defined by the appended claims. The detailed description that follows is written from the point of view of a control systems company, so it is to be understood that generally the concepts discussed herein are applicable to various subsystems and not limited to only a particular controlled device or class of devices, such as personal area networks that can include mesh networks.

Reference throughout the specification to "one embodiment" or "an embodiment" means that a particular feature, structure, or characteristic described in connection with an embodiment is included in at least one embodiment of the embodiments. Thus, the appearance of the phrases "in one embodiment" on "in an embodiment" in various places throughout the specification is not necessarily referring to the same embodiment. Further, the particular feature, structures, or characteristics may be combined in any suitable manner in one or more embodiments.

LIST OF REFERENCE NUMBERS FOR THE ELEMENTS IN THE DRAWINGS IN NUMERICAL ORDER

The following is a list of the major elements in the drawings in numerical order.

100 Mesh Network
102 Coordinator Node
104 Full Function Device (FFD) Node
106 Reduced Function Device (RFD) Node
110 Diagnostic Node
200 Personal Area Network (PAN)
202 Gateway Device
204 Lighting Control
206 Thermostat
208 Wireless Expander
210 Diagnostic Console
250 Control System
251 Control Processor (Processor)
252 Lighting Control
253 Control System Bus (Ethernet, USB, RS232, Cresnet)
254 Sky Light Control
256 Drapery Control
258 Shade Control
260 HVAC Control
262 Audio Video Elements
264 Household Appliance Elements
266 Security Elements
302 Radio Frequency Transceiver Interface
303 Antenna
304 Power-On Indicator
306 Network Indicator
308 Activity Indicator
310 Acquire Button
312 Setup Button
314 LAN Power Over Ethernet (POE) Connector
316 CrestNet Interface
318 Processor
320 Processor Internal Memory
322 Executable Program/Application (App)
402 Integrated Display/Touch-Screen (laptop/tablet etc.)
404 Internal Data/Command Bus (Bus)
410 Universal Serial Bus (USB) Port
416 Hard Disk Drive (HDD)
418 Read-Only Memory (ROM)
420 Random Access Memory (RAM)
432 Processor Board/PC Internal Memory (Internal Memory)
434 Flash Drive Memory
442 Wi-Fi Transceiver
444 BlueTooth (BT) Transceiver
446 Near Field Communications (NFC) Transceiver
452 Antenna
456 Universal Serial Bus (USB) Cable
458 Ethernet Cable (CATS)
502 Diagnostic Console Wired Input/Output Interface
504 Diagnostic Console Display/Touchscreen
506 Diagnostic Console Buttons
508 Diagnostic Console Keypad
510 Diagnostic Console Microphone
512 Diagnostic Console Speaker
602 Mode Button

604 Fan Button
606 Display
608 Up Button
610 Down Button
700*a,b* First and Second Link Debug Table
702 Link Debug Data Reporting Packet
800 Method for Diagnosing Communication Links in a Personal Area Network
802-816 Method Steps for Method **800**

The following is a list of the acronyms used in the specification in alphabetical order.

ALQI Application Link Quality Indicator
APS Application Support Sublayer
APSFR APS Layer Transmit Failure Rate
APSRR APS Layer Transmit Retry Rate
ASIC Application Specific Integrated Circuit
AV Audio/Visual
BFEXH Buffer Exhaustion on a Network Device
BIOS Basic Input/Output System
BT Blue Tooth
CD Compact Disk
CPU Central Processing Unit
CSC Crestron Shade Control
DVD Digital Versatile Disk
EEPROM Electrically Erasable Programmable Read Only Memory
EUID Effective User Identification
FFD Full Function Device
FPGA Field Programmable Gate Array
GAN Global Area Network
H Hex
HDD Hard Disk Drive
HVAC Heating, Ventilation, Air Conditioning
I/O Input/Output
IEEE Institute of Electrical and Electronic Engineers
ISA International Society of Automation
ISM Industrial, Science, and Medical
LAN Local Area Network
LED Light Emitting Diode
LQI Link Quality Indicator
LR-WPAN Low Rate Personal Area Network
MAC Media Access Control
MACFR MAC Layer Transmit Failure Rate
MACRR MAC Layer Transmit Retry Rate
MTORR RX Map-To-One Route Requests Received
NFC Near Field Communications
NIC Network Interface Card
NTRNVR Network Device's Neighbor Table Turnover
PAN Personal Area Network
PAN ID Personal Area Network Identification
PHY Physical Layer
RAM Random Access Memory
RF Radio Frequency
RFD Reduced Function Device
RISC Reduced Instruction Set
ROM Read Only Memory
RSSI Received Signal Strength Indication
RXCORR Received Corrupt Message Rate
TBMT Time Between Map-to-one Route Requests
TSID Transport Stream Identification
TSID Transmitter Subscriber ID/Crestron Touch Settable ID
USB Universal Serial Bus
UWB Ultra-Wide Band
WAN Wide Area Network
ZLQI Zigbee (stack layer) Link Quality Indicator

MODE(S) FOR CARRYING OUT THE INVENTION

Aspects of the embodiments provide a wireless personal area network (PAN) operating according to one of a plurality of wireless communications protocols and comprising one or more network devices. The PAN is configured for employing a debugging or diagnostic apparatus and process that provides extended communication diagnostic data. The PAN diagnostic process and apparatus provides several sources of information to provide installers a more detailed picture as to why communication problems can exist in the PAN.

According to further aspects of the embodiments, high (i.e. application-detected) and low (i.e. stack detected) information are combined to determine in a more effective and efficient manner the quality of a communication link in a PAN. Typical solutions for 15.4/Zigbee networks, as provided by stack software, are forced to use the lowest layer only (i.e., the media access control (MAC) and physical (PHY) layers), which greatly restrict the information available for analyzing and diagnosing communication link errors. As those of skill in the art can appreciate, this restriction is the result of an Institute of Electrical and Electronic Engineers (IEEE) 15.4 requirement for ascertaining link quality that the physical (PHY) layer has to be able to determine the value. This restriction not only severely limits the information available in the Zigbee LQI ("ZLQI") determination, but also means the quality of the ZLQI determination is poor, and it is based on only one data source (the received frame error rates). As those of skill in the art can appreciate, the ZLQI measurement is a characterization of the strength and/or quality of a received packet. The measurement may be implemented using receiver ED, a signal-to-noise ratio estimation, or a combination of these methods. The use of the ZLQI result by the network or application layers is not specified in this standard.

The ZLQI measurement can be performed for each received packet, and the result can be reported to the MAC sublayer using PD-DATA.indication as an integer ranging from 0x00 to 0xff. The minimum and maximum LQI values (0x00 and 0xff) should be associated with the lowest and highest quality IEEE 802.15.4 signals detectable by the receiver, and LQ values in between should be uniformly distributed between these two limits. At least eight unique values of LQ can be used. Using PHY-layer only in the ZLQI limits the usability of the ZLQI value. The method used in the prior art versions (at least in earlier hardware/stack versions) is based on the receiver correlation error on incoming data—data can be received correctly but with errors in it, the more errors, the worse the ZLQI. The problems with this method are poor rangeability (very few effective "quality" levels), and incomplete data application. In addition, once error rates become too high to successfully decode a packet, that event fails to become factored into the ZLQI; as a result, intermittent links do not properly drag down the ZLQI value because the "failed" receptions (that totally fail decoding) do not get applied to the ZLQI value.

According to aspects of the embodiments, communication link information is automatically reported by the devices periodically and collected by PAN **100**. PAN **100** then makes available the collected communication link information, which is organized by gateway console command, thereby allowing conditions in all areas of the network to be quickly assessed. According to aspects of the embodiments, one

method to retrieve the data is by console command in the gateway/mc3; in this method, the user can open a software tool such as Crestron Toolbox to get a "console window" in which simple text-based commands can be sent and text responses received. This data can be retrieved by a Simpl+ program to be formatted and sent to conventional user interfaces such as TouchPanel, which is a type of application that can be used in the system described herein according to aspects of the embodiments. According to further aspects of the embodiments, other interfaces for control applications, such as Pyng, can be used in which data can be reported to the Pyng application without having to issue console commands and parse responses. According to further aspects of the embodiments, a derived link quality indicator (LQI) metric can be generated. The LQI is based on an aggregation of the collected communication link information. According to still further aspects of the embodiments, a second derived link quality indicator can also be generated, and is referred to as an "application LQI" (ALQI). The ALQI provides a relatively quick assessment of communication quality. A detailed discussion of the determination of both the LQI and ALQI is made below, in reference to FIG. **7**. In addition, an interpretation of the determined values of LQI and ALQI is also described in detail below.

Aspects of the embodiments are described herein in the context of a PAN (e.g. PAN **200**) operating with proprietary wireless network protocols, such, e.g., the infiNET EX protocols, and wherein the PAN comprises one or more of a lighting device, a drapery device, a shade device, an HVAC device (such as a thermostat), a control processor, and a wireless gateway as the network devices of the PAN, but is not limited thereto. Those of skill in the art can appreciate that in accordance with currently available personal area network communications technology, virtually any household device, as well as virtually any commercially used device, can be interfaced with one or more communications devices that can make the interfaced device part of the PAN. Consequently, the above list is not limiting, and should not be taken in such regard. Any device configured for communicating on a wireless network such as PAN can serve as the joining device. For example, any one or an AV device, a security device, a household appliance, a portable communication device, among virtually any household or commercially used devices can be a network device in the PAN.

FIG. **2** illustrates wireless PAN **200** (PAN **200**) operating according to wireless protocols (e.g., infiNET) and connected to control system **250** according to aspects of the embodiments. PAN **200** comprises one or more network devices such as a wireless gateway (gateway device) **202** acting as a PAN coordinator (e.g., coordinator node **102**) and communicating with a variety of other network devices including a plurality of lighting controls **204***a-d*, thermostat **206**, wireless expander **208** and diagnostic console **210**, each of which act as either FFD nodes **104**, or RFD nodes **106** (with the exception of diagnostic console **210**, discussed in greater detail below). According to an aspect of the embodiments, PAN **200** can be operating according to a proprietary communication protocol, such as infiNET EX protocols. According to this embodiment, gateway device **202** can be a CEN-RFGW-EX infiNET EX wireless gateway available from Crestron Electronics, Inc. of Rockleigh, N.J. Lighting controls **204** can comprise infiNET EX wall box controls such as CLW-DIMEX-P and CLW-DIMSWEX-P and a dimmer such as a CLW-LDIMEX infiNET lamp dimmer, all available from Crestron Electronics, Inc. of Rockleigh N.J. Wireless expander **208** can be a CLW-EXPEX infiNET wireless expander available from Crestron Electronics, Inc. of Rockleigh N.J. Diagnostic console **210** can be embodied in a touch panel form, such as in a MTX-3 Isys™ 2.8" Handheld Wireless Touch Panel w/infiNET EX, which also available from Crestron Electronics, Inc. of Rockleigh N.J.

Those skilled in the art will recognize that additional devices can operate within PAN **200**. Substantially any device capable of communicating on PAN **200**, such as AV controls, HVAC elements, security elements, and other residential or commercial control and monitoring devices can be part of PAN **200** shown in FIG. **1**. For example, PAN **200** can also further comprise HVAC controls, such as a CHV-TSTATEX thermostats available from Crestron Electronics, Inc. of Rockleigh N.J., or shade controls such as Creston shade controls (CSC) that are also available from Crestron Electronics, Inc. of Rockleigh N.J.

Control system **250** can comprise control processor (processor) **251** adapted to control one or more controllable elements of a residential or commercial structure either directly or via intermediate components, and for interfacing with PAN **200** according to aspects of the embodiments. For example, processor **251** can control lighting elements **252**, sky light elements **254**, drapery elements **256**, shade elements **258**, AV elements **262**, household appliance elements **264**, security elements **266**, and HVAC elements **260**. Of the devices **251**, **255**, **254**, **256**, **258**, **260**, **262**, **264**, and **266** can act as either FFD nodes **104** or RFD nodes **106**.

FIG. **3** illustrates a wireless gateway device (gateway device) **202** according to an aspect of the embodiments. Gateway device **202** is adapted for acting as PAN coordinator node **102** and managing network configurations. Additionally, gateway device **202** is further adapted to serve as interface (or gateway) for PAN **200** to communicate with control system **250** according to aspects of the embodiments. Gateway device **202** can be a CEN-RFGW-EX wireless gateway available from Crestron Electronics, Inc. The CEN-RFGW-EX wireless gateway is a two-way RF gateway\transceiver designed to enable communications and management for a complete infiNET EX wireless network of dimmers, keypads, remotes and other devices. The CEN-RFGW-EX wireless gateway links the infiNET EX network to a Crestron control system via a wired connection such as Ethernet or Cresnet. Up to one hundred infiNET EX dimmers, switches, keypads, thermostats, and other devices can be linked to a control system, such as control system **250**, via a single CEN-RFGW-EX gateway adapted to perform as gateway device **202**. Additional gateways can be installed to support more devices with up to sixteen gateways possible in a complete system. Wireless expanders **208** can also be added wherever needed to extend the network by filling in gaps between devices.

Gateway device **202** can include at least one central processing unit (CPU) **318**. For example, CPU **318** can represent one or more microprocessors, and the microprocessors can be "general purpose" microprocessors, a combination of general and special purpose microprocessors, or application specific integrated circuits (ASICs). Additionally or alternatively, CPU **318** can include one or more reduced instruction set (RISC) processors, video processors, or related chip sets. CPU **318** can provide processing capability to execute an operating system, run various applications, and/or provide processing for one or more of the techniques described herein. Such an Applications that can run on the gateway device **202** can include, for example, software for initiating PAN **200**, communicating on control system **250**, and PAN communications link diagnostic software/application (App) **322**.

FIGS. **3** and **4** illustrate gateway device **202** suitable for use to implement method **500** for diagnosing communication links within a personal area network according to an embodiment. Gateway device **202** comprises, among other items, a shell/box/container, integrated indicators **304**, **306**, **308**, internal data/command bus (bus) **404**, processor board/PC internal memory (internal memory) **432**, and one or more processors **408** with processor internal memory **406** (which can be typically read only memory (ROM) and/or random access memory (RAM)). Those of ordinary skill in the art can appreciate that in modern PC systems, parallel processing is becoming increasingly prevalent, and whereas a single processor would have been used in the past to implement many or at least several functions, it is more common currently to have a single dedicated processor for certain functions (e.g., digital signal processors) and therefore could be several processors, acting in serial and/or parallel, as required by the specific application. Gateway device **202** further comprises multiple input/output ports, such as universal serial bus ports **410**, LAN PoE **314**, CrestNet interface **316**, among other types. Gateway device **202** still further includes wireless communication apparatus, such as one or more of the following: Wi-Fi transceiver **442**, BlueTooth (BT) transceiver **444**, near field communications (NFC) transceiver **446**, and RF transceiver interface **302**, and antenna **303**.

Internal memory **432** itself can comprise hard disk drive (HDD) **416** (these can include conventional magnetic storage media, but, as is becoming increasingly more prevalent, can include flash drive memory **434**, among other types), read-only memory (ROM) **418** (these can include electrically erasable (EE) programmable ROM (EEPROMs), ultraviolet erasable PROMs (UVPROMs), among other types), and random access memory (RAM) **420**. Usable with universal serial bus (USB) port **410** is flash drive memory **434**. Each of the memory storage devices, or the memory storage media (**406**, **418**, **420**, **434**, among others), can contain all or some of executable software programming code or application (application, or "App") **322**, which can implement part or all of the portions of method **500** described herein. In addition, any of the memory devices can buffer or cache data. In buffering or caching data related to operations of processor **318**, memory **320**, **432** can store data associated with applications running on the gateway device **202**.

As mentioned above, gateway device **202** further comprises a plurality of wireless transceiver devices, such as Wi-Fi transceiver **442**, BT transceiver **444**, NFC transceiver **446**, RF transceiver device **302**, and antenna **303**. While each of Wi-Fi transceiver **442**, BT transceiver **444**, NFC transceiver **446**, and RF transceiver device **302** has their own specialized functions, each can also be used for other types of communications, such as accessing a cellular service provider (not shown), accessing the internet, among other types communications and data/voice transfers/exchanges, as known to those of skill in the art. Each of Wi-Fi transceiver **442**, BT transceiver **444**, NFC transceiver **446**, and RF transceiver device **302** includes a transmitting and receiving device, and a specialized antenna, although in some instances, one antenna can be shared by one or more of Wi-Fi transceiver **442**, BT transceiver **444**, NFC transceiver **446**, and RF transceiver device **302**. Alternatively, one or more of Wi-Fi transceiver **442**, BT transceiver **444**, NFC transceiver **446**, and RF transceiver device **302** will have a specialized antenna, such as RF transceiver device **302** to which is electrically connected at least one antenna **303**. Gateway device **202** can also be part of a larger network configuration as in a global area network (GAN) (e.g., the internet), which ultimately allows connection to various landlines.

According to further embodiments, integrated displays **304**, **306**, and **308**, and acquire and setup buttons **310**, **312** respectively, as well as CrestNet interface **316** and LAN PoE interface **314** can provide a means for a user to enter commands, data, digital, and analog information into gateway device **202**, and to determine status as well. Integrated displays **304**, **306**, and **308** can be used to show visual representations of acquired data, and the status of applications that can be running, among other things.

Bus **404** provides a data/command pathway for items such as: the transfer and storage of data/commands between processor **318**, Wi-Fi transceiver **442**, BT transceiver **444**, NFC transceiver **446**, RF transceiver device **302**, integrated displays **304**, **306**, and **308**, buttons **310**, **312**, USB port **410**, LAN PoE interface **314**, and internal memory **432**. Through bus **404**, data can be accessed that is stored in internal memory **432**. Processor **318** can send information for visual display to one or more of displays **304**, **306**, and **308**, and the user can send commands to App **322** that can reside in processor internal memory **320** of processor **318**, or any of the other memory devices (**416**, **418**, **420**, **434**).

Gateway device **202**, and either processor internal memory **320** or internal memory **432**, can be used to implement method **500** for diagnosing communication links in a personal area network according to an embodiment. Hardware, firmware, software or a combination thereof may be used to perform the various steps and operations described herein. According to an embodiment, App **322** for carrying out the above discussed steps can be stored and distributed on multi-media storage devices such as flash drive memory **434** (described above) or other form of media capable of portably storing information. Storage media **434** can be inserted into, and read by devices such as USB port **410**.

As also will be appreciated by one skilled in the art, the various functional aspects of the embodiments may be embodied in a wireless communication device, a telecommunication network, or as a method or in a computer program product. Accordingly, the embodiments may take the form of an entirely hardware embodiment or an embodiment combining hardware and software aspects. Further, the embodiments may take the form of a computer program product stored on a computer-readable storage medium having computer-readable instructions embodied in the medium. Any suitable computer-readable medium may be utilized, including hard disks, compact disk (CD)-ROMs, digital versatile discs (DVDs), optical storage devices, or magnetic storage devices such a floppy disk or magnetic tape. Other non-limiting examples of computer-readable media include flash-type memories or other known types of memories.

Further, those of ordinary skill in the art in the field of the embodiments can appreciate that such functionality can be designed into various types of circuitry, including, but not limited to field programmable gate array structures (FPGAs), application specific integrated circuitry (ASICs), microprocessor based systems, among other types. A detailed discussion of the various types of physical circuit implementations does not substantively aid in an understanding of the embodiments, and as such has been omitted for the dual purposes of brevity and clarity. However, as well known to those of ordinary skill in the art, the systems and methods discussed herein can be implemented as discussed, and can further include programmable devices.

Such programmable devices and/or other types of circuitry as previously discussed can include a processing unit, a system memory, and a system bus that couples various system components including the system memory to the processing unit. The system bus can be any of several types of bus structures including a memory bus or memory controller, a peripheral bus, and a local bus using any of a variety of bus architectures. Furthermore, various types of computer readable media can be used to store programmable instructions. Computer readable media can be any available media that can be accessed by the processing unit. By way of example, and not limitation, computer readable media can comprise computer storage media and communication media. Computer storage media includes volatile and non-volatile as well as removable and non-removable media implemented in any method or technology for storage of information such as computer readable instructions, data structures, program modules or other data. Computer storage media includes, but is not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, DVD, or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by the processing unit. Communication media can embody computer readable instructions, data structures, program modules or other data in a modulated data signal such as a carrier wave or other transport mechanism and can include any suitable information delivery media.

The system memory can include computer storage media in the form of volatile and/or nonvolatile memory such as ROM and/or RAM. A basic input/output system (BIOS), containing the basic routines that help to transfer information between elements connected to and between the processor, such as during start-up, can be stored in memory. The memory can also contain data and/or program modules that are immediately accessible to and/or presently being operated on by the processing unit. By way of non-limiting example, the memory can also include an operating system, application programs, other program modules, and program data.

The processor can also include other removable/non-removable and volatile/nonvolatile computer storage media. For example, the processor can access a hard disk drive that reads from or writes to non-removable, nonvolatile magnetic media, a magnetic disk drive that reads from or writes to a removable, nonvolatile magnetic disk, and/or an optical disk drive that reads from or writes to a removable, nonvolatile optical disk, such as a CD-ROM or other optical media. Other removable/non-removable, volatile/nonvolatile computer storage media that can be used in the operating environment include, but are not limited to, magnetic tape cassettes, flash memory cards, digital versatile disks, digital video tape, solid state RAM, solid state ROM and the like. A hard disk drive can be connected to the system bus through a non-removable memory interface such as an interface, and a magnetic disk drive or optical disk drive can be connected to the system bus by a removable memory interface, such as an interface.

The embodiments discussed herein can also be embodied as computer-readable codes on a computer-readable medium. The computer-readable medium can include a computer-readable recording medium and a computer-readable transmission medium. The computer-readable recording medium is any data storage device that can store data which can be thereafter read by a computer system. Examples of the computer-readable recording medium include ROM, RAM, CD-ROMs and generally optical data storage devices, magnetic tapes, flash drives, and floppy disks. The computer-readable recording medium can also be distributed over network coupled computer systems so that the computer-readable code is stored and executed in a distributed fashion. The computer-readable transmission medium can transmit carrier waves or signals (e.g., wired or wireless data transmission through the Internet). Also, functional programs, codes, and code segments to, when implemented in suitable electronic hardware, accomplish or support exercising certain elements of the appended claims can be readily construed by programmers skilled in the art to which the embodiments pertains.

The disclosed embodiments provide a system and method for diagnosing communication links in a personal area network. It should be understood that this description is not intended to limit the embodiments. On the contrary, the embodiments are intended to cover alternatives, modifications, and equivalents, which are included in the spirit and scope of the embodiments as defined by the appended claims. Further, in the detailed description of the embodiments, numerous specific details are set forth to provide a comprehensive understanding of the claimed embodiments. However, one skilled in the art would understand that various embodiments may be practiced without such specific details.

As described above, gateway device **202** can also include nonvolatile storage. The nonvolatile storage can represent any suitable nonvolatile storage medium, such as HDD **416** or nonvolatile memory, such as Flash memory. Being well-suited to long-term storage, the nonvolatile storage can store data files, software (e.g., for implementing functions on gateway device **202**), and wireless connection information (e.g., information that can enable media device to establish a wireless connection such as a telephone connection).

One or more network interfaces can provide connectivity for gateway device **202**. The network interfaces can represent, for example, one or more network interface cards (NIC) or a network controller. In certain embodiments, the network interface can include a PAN interface, such as a two-way RF transceiver **302**. RF transceiver **302** can provide capabilities to network with, for example, a Bluetooth® network, an IEEE 802.15.4 (e.g., ZigBee) network, or an ultra-wideband network (UWB). As can be appreciated, the networks accessed by RF transceiver **302** can, but do not necessarily, represent low power, low bandwidth, or close range wireless connections. RF transceiver **302** can permit one electronic device to connect to another local electronic device via an ad-hoc or peer-to-peer connection, as is evident of Mesh networks.

The network interface can also include one or more local area network (LAN) interfaces, such as LAN PoE interface **314**. The LAN interface can represent an interface to a wired Ethernet-based network, but can also represent an interface to a wireless LAN, such as an IEEE 802.11x wireless network. The range of the LAN interface can generally exceed the range available via RF transceiver **302**. Additionally, in many cases, a connection between two electronic devices via the LAN interface can involve communication through a network router or other intermediary device.

According to aspects of the embodiments, LAN PoE interface **314** is an 8-wire RJ-45 female connection with two LED indicators. Cresnet interface **316** is a 4-pin 3.5 mm detachable terminal block providing an interface for Cresnet proprietary communications on a LAN. LAN PoE interface can be adapted to receive both an electric power signal and an information signal from a control network. For example,

LAN PoE interface can be connected through category **5** cable (CAT **5**) to a LAN that contains both a power supply and multiple control points and signal generators. Through LAN PoE interface or Cresnet interface **316**, gateway device **202** can interface with control system **250**. For example, gateway device **202** can communicate with processor **251**, such as a PRO3 available from Crestron Electronics, Inc.

Referring back again to FIG. **3**, gateway device **202** comprises one or connectors, indicators and interface buttons. Gateway device **202** comprises RF transceiver interface **302** to which is attached antenna **303**. Gateway device **202** further comprises power-on LED (PWR) **304**, network LED (NET) **306** and activity LED **308**. Power LED **304** is an LED indicating that operating power is being supplied whether from the Cresnet network or the PoE connection. Network LED **306** indicates communication with the Cresnet system. Activity LED **308** indicates wireless communications, such as with the elements of the wireless PAN **200**.

Gateway device **202** further comprises acquire button **310** and setup button **312**. Acquire button **310** and setup button **312** are recessed push buttons each with an indicator LED. Acquire button **310** is employed to configure communication with PAN **200** and setup button **312** is employed to configure communication with control system **250** according to further aspects of the embodiments. Discussion will now be made of how to configure communications between one or more nodes within PAN **200**, and communications between PAN **200** and control system **250** according to aspects of the embodiments.

Gateway device **202** can be placed in acquire mode by pressing acquire button **310**. The associated LED will indicate that gateway device **202** is in acquire mode when illuminated. Once gateway device **202** (i.e., as the coordinator node **102** of PAN **200**) is placed in acquire mode, any device (e.g., any of the household or commercial devices that can become PAN **200** nodes, e.g., the "joining devices") that is within range of gateway device **202** or which can be brought into range, will receive a notification from coordinator node **102**, and will internally go to an acquire mode so that it can be acquired by gateway device **202**. The joining device will be automatically acquired by gateway device **202**. Pressing acquire button **310** a second time will cause Gateway device **202** to exit acquire mode as indicated by an LED associated or part of acquire button **310**, which will then extinguish.

PAN **200** can further comprise diagnostic console **210**, as shown in FIG. **2** and in greater detail in FIG. **5**. As described above, the interface for joining devices such as thermostat **206** to PAN **200** can be limited in functionality and not capable of providing adequate information to an installer of PAN **200**. Diagnostic console **210** can be a dedicated device or can represent an added functionality of an existing device, such as a tablet computer or smart phone (i.e., meaning that a special program or App has been added to the tablet or smart phone). According to one non-limiting example, diagnostic console **210** is being shown in FIG. **2** as being incorporated as, or part of, a separate device. As such, according to this non-limiting example, one or more network interfaces, e.g., gateway device **202**, can provide connectivity for diagnostic console **210**. According to aspects of the embodiments, gateway device **202** manages and communicates with one PAN **200**, although according to further aspects of the embodiments, gateway device **202** can manage and communicate with more than one PANs **200**. The network interfaces can represent, for example, one or more NICs, or a network controller. According to aspects of the embodiments, diagnostic console **210** operates to provide available debugging data to one or more users, regardless of the location of the user (e.g., the user can access diagnostic console **210** via one or more other networks, such as, but not limited to, wireless cellular networks, the internet, intranets (such as CresNet), among many other network types). Consequently, as described, diagnostic console **210** does not affect any aspect of network operation according to one non-limiting aspect of the embodiments. In other aspects, diagnostic console **210** can affect operation. According to further aspects of the embodiments, diagnostic console **210** can include such functionality as generating channel changes in response to system-wide communication problems, or provide troubleshooting steps/items of concern to the installer. In certain embodiments, the network interface can include a PAN interface, such as a two-way RF transceiver **302**. The PAN interface can provide capabilities to network with, for example, a Bluetooth network, an IEEE 802.15.4 (e.g., ZigBee) network, or a UWB. As can be appreciated by those of skill in the art, the networks accessed by the PAN interface can be characterized as exhibiting low power, low bandwidth, or close range wireless connections, among other features. The PAN interface can permit one electronic device to connect to another local electronic device via an ad-hoc or peer-to-peer connection.

Diagnostic console **210** can also include wired input/output (I/O) interface **502** for a wired interconnection between itself and another electronic device, such as a joining device. The wired I/O interface can represent, for example, a universal serial bus (USB) port or an IEEE 1394 or FireWire® port, but can also represent a proprietary connection.

The wired interface of diagnostic console **210** can be further adapted to receive diagnostic information from the joining device, and diagnostic console **210** can further include display/touchscreen **504** for displaying the diagnostic information. For example, diagnostic console **210** can receive and display the table of discovered networks from the joining device. As those of skill in the art can appreciate, when a device joins a network such as PAN **200**, or a mesh network, the joining device creates a table of networks that the joining device did join, or could have joined. According to further aspects of the embodiments, diagnostic console **210** does not typically perform its own commissioning process; however, as those of skill in the art can appreciate, according to further aspects, the commissioning process can be incorporated within diagnostic console **210**. For example, diagnostic console **210** can be provided with such commissioning tools and can therefore generate its own table of entries for up to four discoverable networks (in an "acquire scan" application) that it may wish to join with. According to further aspects of the embodiments, gateway device **202** does not typically control what devices join to it, although according to further aspects of the embodiments such discrimination can be built-in or added as function to gateway device **202**. According to further aspects of the embodiments, diagnostic console **210** can provide network technicians with the ability to diagnose problem installations and help them understand the wireless environment in which devices are placed. Such features could be implemented by allowing diagnostic console the ability to join an existing (or different) network, as opposed to the gateway it is currently attached to.

Diagnostic console **210** can further comprise one or more user interface elements for receiving and displaying information. For example, the one or more user interface ele-

ments can comprise one or more buttons **506***a-e*, keypad **508**, microphone **510**, speaker **512**, and indicator lights/lamps (not shown).

FIG. **6** illustrates thermostat **206** for use in PAN **200** as shown in FIG. **2** according to aspects of the embodiments. Thermostat **206** can be a CHV-TSTATEX thermostat available from Crestron Electronics, Inc. of Rockleigh, N.J. The CHV-TSTATEX thermostat can operate on PAN **200** by joining it and being acquired by gateway device **202** as described above.

Thermostat **206** is adapted to control heating and cooling of baseboard, forced air, heat pump and dual fuel heat pump HVAC systems. Thermostat **206** is further configured for communicating on PAN **200**.

Thermostat **206** can include at least one CPU. For example, the CPU can represent one or more microprocessors, and the microprocessors can be "general purpose" microprocessors, a combination of general and special purpose microprocessors, or ASICs. Additionally, or alternatively, the CPU can include one or more RISC processors, video processors, or related chip sets. The CPU can provide processing capability to execute an operating system, run various applications, and/or provide processing for one or more of the techniques described herein. Applications that can run on thermostat **206** can include, for example, software for communicating on PAN **200** and controlling one or more HVAC elements **260**.

A main memory can be communicably coupled to the CPU, which can store data and executable code. The main memory can represent volatile memory such as RAM, but can also include nonvolatile memory, such as ROM or flash memory. In buffering or caching data related to operations of the CPU, the main memory can store data associated with applications running on thermostat **206**.

Thermostat **206** can also include nonvolatile storage. The nonvolatile storage can represent any suitable nonvolatile storage medium, such as a HDD or nonvolatile memory, such as flash memory. Being well-suited to long-term storage, the nonvolatile storage can store data files, software (e.g., for implementing functions on thermostat **206**), and wireless connection information (e.g., information that can enable thermostat **206** to establish a wireless connection).

As discussed above, thermostat **206** can also include a wired input/output (I/O) interface for a wired interconnection between itself and another electronic device. The wired I/O interface can represent, for example, a USB port or an Institute of Electrical and Electronic Engineers (IEEE) 1394 or FireWire® port, but can also represent a proprietary connection.

Thermostat **206** further comprises LCD display **606** and one or more user interface buttons **602**, **604**, **608**, and **610**. Thermostat **206** can be placed in acquire mode via a series of menus displayed on LCD display **606**. Selections are made via the one or more user interface buttons **602**, **604**, **608**, and **610**. For example, the user can enter a setup mode by pressing and holding fan button **604**. While fan button **604** is being held, the user can then simultaneously press and hold both up button **608** and down button **610** for some period of time (e.g., five seconds) to enter setup mode. Once in setup mode, thermostat **206** can be placed in acquire mode by actuating mode button **602** twice and then selecting either up button **608** or down button **610**. An LED display can then indicate that thermostat **206** is performing the scan and acquire process by flashing a " - - - " on display **606**. Upon completion of the scan and acquire process, LED display **606** will indicate a "00" if thermostat **206** has successfully been acquired or an "E#" to indicate an error where # indicates an error code. For example, the error codes "E**1**", "E**2**" and "**3**" can correspond to the following conditions, respectively: no gateway found, gateway found but device rejected, and radio operation not available.

As described above, one or more devices in PAN **200** can collect and report data, and one or more devices can analyze this data to diagnose issues in PAN **200**. According to aspects of the embodiments, and from hereon on, unless specifically described otherwise, "data" refers to a link debug data reporting packet. For example, network devices that can collect and report data include a CLWI-KPLEX In-Wall Wireless Lighting Keypad, and a CLWI-KPLEX-BATT In-Wall Wireless Lighting Keypad, both of which are available from Crestron Electronics, Inc., and both of which can be considered to be non-limiting examples of joining devices of PAN **200**. The MC3 control processor and DIN-AP3MEX DIN rail 3 series automation processor can be considered as representative, but non-limiting examples of processor **251**, and can analyze data from other devices in PAN **200**. CEN-RFGW-EX gateway device, which can be considered a representative, but non-limiting example of gateway device **202**, can also analyze data. All three of these devices are available from Crestron Electronics, Inc.

Routers and always-on devices report the data to the gateway as an average time between map-to-one route requests received (MTORR RXs) response. Routers, as those of skill in the art can appreciate, can be described as a device that participates in routing or moving a message from one place to another. Advantageously, this minimizes bandwidth consumption as the incremental increase in busy time (by adding this data) is very small. According to aspects of the embodiments, routers are defined as any node within PAN **200** that can route a message. An "always-on" device is a node that can hear any over-the-air messages, although it does not necessarily need to have to response to them or participate in routing them. According to further aspects of the embodiments, non-battery operated devices are typically designated as "always-on" and routers, while battery operated devices are typically designated as "end devices." An "end devices," according to aspects of the embodiments, does not have the responsibility and/or capability of routing messages, as a router-type device does. According to still further aspects of the embodiments, "busy time" is that time in which the device is not able to communicate because the energy level of the communication path is above some level, meaning that either noise is present, or other devices within PAN **100** are communicating, and the instant device would not be able to be heard above the existing energy levels.

Certain end devices, such as battery powered devices, report the data at times that are convenient for the application specific to the device in a manner that minimizes power impact. Such applications can include, by way of non-limiting example, software/firmware that implements one or more specific functionalities of the devices such as a button/LED user interface code in a keypad. According to further aspects of the embodiments, applications can include that software/firmware created by Crestron to implement the specific functionalities of the respective devices. In general this means it is reported when the application already needs to talk, in a manner that does not increase the over-the-air frame count. In an embodiment, a device such as the CLWI-KPLEX-BATT can piggyback the data on release edges of any button press. In other embodiments, battery powered network devices can piggyback the data on other events. For example, an occupancy sensor can piggyback the data on occupancy and vacancy events during the analog output for timeout. According to further aspects of the

embodiments, multiple messages can be concatenated into one over-the-air packet so the App message (i.e. button press) and the link debug data packet can be sent in one transmit event.

FIG. **8** illustrates a flowchart of method **800** for diagnosing communication links in PAN **200** according to aspects of the embodiments, and FIGS. **7**A, **7**B, and **7**C illustrate tables that show a plurality of link debug table used in method **800** for diagnosing communication links in PAN **200** according to aspects of the embodiments. Referring back to FIG. **2**, gateway device **202** can obtain and report link debug data by issuing a command to one or more network devices. The command issued to the network device obtains statistical information about the state of communication for the network device in relation to both their local environment (e.g., between the network device to which the command was issued and other network devices or devices that communicate with the network device (e.g., a wireless light switch can communicate wireless with gateway device **202** and the light(s) it controls)) and end-to-end communications between gateway device **202** and the network device. Once one or more of such commands have been issued, a table can be assembled by gateway device **202** such as that shown in FIGS. **7**A, **7**B, and **7**C that reports the link debug communication data for the entire PAN **200**.

Referring now to FIGS. **7**A, **7**B, and **7**C, it can be seen that there are six devices reporting communication link debug data. The six nodes have the following network identities (as shown in the "ID" column): 12, 13, 14, 20, 21, 60, and 61, and have been given, respectfully, node addresses (as shown in the "NODE" column), D47F(H), 447E(H), 6C80(H), A957(H), F53C(H), and 10A7(H) (wherein "(H)" denotes a hexadecimal number). For each network device (i.e., for each row), there are twenty three fields. The fields and expected range of values are shown and defined below.

The first column, for FIGS. **7**A, **7**B, and **7**C (so that the values of the fields can be more readily related to the node to which they pertain), pertains to the identification (ID) field. The ID field is the network identification of the network device responding to the communication link debug data command. The second column is the Node field. The Node field is the short address of the node. Using a Node field assists in identifying devices if there are duplicate ID's in the network. Due to screen width and formatting difficulty, it would not be practical to report a transmitter subscriber ID (TSID; according to further aspects of the embodiments, the TSID is the Crestron touch-settable ID or product serial number of the device) or effective user ID (EUID). The Node field value will always be unique within the particular network. As those of skill in the art can appreciate, the EUID is also known as the IEEE EUI-64.

The stack derived Zigbee LQI value (ZLQI) is an LQI based on error rates in decoding received signals. Because of limitations in how certain standards' bodies require this value to be derived, it has been shown to offer limited value in debugging practical systems. According to aspects of the embodiments, fields **16-23** are used to generate the ALQI value of field **3**, which offers a better tool for debugging the communication links of PAN **200**. The values reported in field **8-15** can be used for informational purposes, or to capture long term trends in operation, so that installers or other users/operators of PAN **200** can verify overall system operation at later times, over longer periods of time. The values of fields **8-15**, the reported ALQI value of field **3**, and the ZLQI value can all be compared and various weights applied, over various and predetermined periods of time in order to discern the overall status of PAN **200** according to further aspects of the embodiments.

Use of ALQI, as described in greater detail below, provides an indicator of a certain link "health." ALQI provides either or both of a pass/fail evaluation of the status of a link, as well as a relative indicator of the link health (such as, but not limited to, "green" (all okay), "yellow" (some, but not overwhelming problems), and "red" (there are significant issues that need to be addressed as soon as practically possible). ALQI provides a substantially quicker overview of the status of a link than can be ascertained from evaluating each of the status indicators.

Field **5** reports the cost field (COST). The cost value is a stack-derived parameter indicating the difficulty in delivering messages to the destination (in this case, from the network device to gateway device **202**). The values of the COST field cover a range of {0,7}, with {0} being preferable. The COST field value totalizes the probability of delivery failure along each node in the path to the destination. On solid networks, it is not unusual to see a COST value of {0} with multiple hops. COST values over {2} generally indicate sustained communication problems, which will likely show up in other parameters in the listing.

Field number **6** reports the TBMT field, or the average time between map-to-one route requests received (MTORR RXs) in seconds. The TBMT field reports the average number of seconds between receiving MTORR messages. In normal, stable networks, this value can be set to be between about 179 to about 180 seconds; however, other ranges of values of time are possible to use. As those of skill in the art can appreciate, the setting of the average number of seconds for the TBMT field is a balance between bandwidth consumption and route-healing responsiveness. As known to those of skill in the art, even with normal, expected jitter, the measurement of TBMT should be about 180 seconds, or three minutes. Values below this indicate application support sublayer (APS) failures from nodes in the system, as additional MTORRs are used to heal routes when messages fail. Very low values (below 60 seconds) are considered to be a significant problem, as the network is consuming considerable bandwidth trying to reform routes.

Filed number **7** reports the received signal strength indication (RSSI) field. The RSSI field provides the average last-hop RSSI value for the network device. Because network devices only talk to the next network device in the route, the RSSI information is only relevant with regard to the link between a first network device and a second one that the first can communicate with, notwithstanding that the first device can hear several other network devices in the network. The RSSI field value is a moving average and network devices that do not communicate very often (such as battery keypads) can take some time to build up a sensible number in this field.

In regard to fields **8-15**, described below, the values are determined or kept track of since that time in which a re-boot has occurred. That is, beginning upon start up, or a re-start, the number of occurrences of the events are kept track of, and reported as shown in FIG. **7**B.

Field number **8** reports the MAC layer transmit failure rate (MACFR). The MACFR number indicates the percentage of MAC transmission attempts that fail. The device's transport layer is not informed about why a MAC transmission event fails; the causes can include "unable to find clear airspace (busy)" or "the destination does not acknowledge the message." The latter can be the result of interference or too great of a range (e.g., weak signal) between network devices.

Field **9** reports the MAC layer transmit retry rate (MACRR). MACRR indicates the percentage of MAC transmission attempts that are retries. A relatively high MACRR rate is a clear indication that a network device has trouble communicating with its next hop, although other non-application data can contribute to this value (e.g., neighbor table exchanges and other network management operations, among others).

Field **10** reports the APS layer transmit failure rate (APSFR). APSFR is the percentage of APS messages (application messages) that fail to be delivered to gateway device **202**. The APSFR provides a level of confidence that a message will be delivered. APSFR indicates that a message got all the way from one network device's application to the destination's application (in this case, gateway device **202**). As those of skill in the art can appreciate, the significance of "application-to-application" means the format provides confirmation that the message went all the way from the application task that sent it, to the other devices' application task receiving it. This is significant because just getting the other device's MAC layer acknowledge does not prove the device received it.

If there are buffering shortages in that device, the message may get lost when it is handed up through the communication layers, and this will happen after it has already sent out the MAC layer acknowledge. APS messages provide two retries in an attempt to obtain a satisfactory transmission (two retries means three attempts—the first attempt, then two retries). Each APS transmission attempt can trigger many MAC transactions. Due to the robust communication structure with three levels of retry for messaging, it is a fairly serious issue to have a sustained, measurable APS failure rate. Very low bandwidth devices (like battery keypads) that encounter a sporadic failure can result in this effect lingering in its APSFR for some time, but more active/router devices should maintain a value of almost zero in normal conditions.

Field **11** reports the APS layer retry rate (APSRR). The APSRR is the percentage of APS send events that are retries for an existing message. Retries at the APS layer are not uncommon even in stable networks, but this should be viewed with caution. Sustained APS retries rates indicate the MAC layer is regularly having trouble sending messages.

Field **12** reports the amount of corrupt received messages (RXCORR). RXCORR is based on counts of messages that were received by the network device, but are unable to be decoded properly because of bit errors. A RXCORR is generally the result of local interference corrupting the received signal, or a very weak received signal. RXCORR is derived with specific criteria to generate a percentage for convenience in analysis. It does not literally correlate to how many received messages are corrupted. Such criteria cannot be implemented as messages with greater levels of corruption will not be detected by the lower layers anyway.

Field **13** reports the medium busy rate (BUSY). BUSY is the approximate percentage of time that the airspace around a network device is considered "busy" and therefore the device cannot talk. Clear-channel assessment is performed on the network device prior to sending a message in order to increase the likelihood that the destination network device will hear it, as well as for medium sharing purposes. As those of skill in the art can appreciate, BUSY can represent a combination of energy detection (e.g., making sure there is no interference), as well as a determination of frame detection in regard to the IEEE 802.15.4 standards, although other uses of BUSY exist as well, as known to those of skill in the art. That is, the medium is busy for energy, or if any other IEEE 802.15.4 network device is talking (regardless of how weak the signal is for the IEEE 802.15.4 network device). This can be important to consider when partitioning networks in dense settings. Having a significant amount of IEEE 802.15.4/ZigBee networks on a channel can make it very busy, even if the networks are distant and weak. A preliminary assessment can be made by using network devices that support the "scan" console command, and looking for channels that have both low energy and little/no detected networks.

Field **14** reports the amount of turnover in the network device's neighbor table (NTRNVR). Neighbor (i.e., neighboring network devices) status and link quality are used in route determination as well as the propagation of broadcast messages and other network maintenance functions. When the content of a network device's neighbor table is changing often, it is an indication that it has a lot of trouble communicating with its neighbors, they are moving in and out of range, or they are going on- and off-line. Regardless, a high NTRNVR increases the likelihood that the network device's send messages can fail, as it cannot count on the neighbors around it being there when it tries to send data. Like the received message corruption field, this parameter is based on a specific change rate and translated to a percentage for relative comparisons, and does not correlate directly to any easily measurable quantity.

Field **15** reports an indication of buffer exhaustion on a network device (BFEXH). BFEXH indicates that a network device cannot allocate memory for stack operations. Generally this is the result of a routing network device being inundated with too much data, before it can retransmit it and free buffering space. This condition is often the result of MAC send difficulties, such as interference or busy state. When a routing network device loses its buffering ability, it will cause messages to fail for network devices that route through it. Very often a network device that has buffer exhaustion issues will be in the routing path of another network device that shows APS retry/failure rates. As in some of the other metrics, this is a derived value for specific reference criteria, and expressed as percentage for convenience for comparing to other data. Buffer exhaustion is an important qualifier of Node performance and routing ability

According to aspects of the embodiment, the ALQI field is a high-level application value derived from certain ones of the link debug data. The ALQI field is an 8 bit hex number with higher values indicating better quality communications. Because the ALQI value aggregates several sources of data over a shorter period of time, it is more effective at showing problems more quickly than the existing ZigBee LQI (ZLQI) number in use currently (described below).

According to further aspects of the embodiments, ALQI provides a quick, but substantially accurate indicator of communication link status that can be used for troubleshooting systems; the other fields (fields **5-15**) provide more detail on the source of problems, albeit on a slower pace. The ALQI field can be updated by the network device on every message it sends; consequently, the ALQI field is therefore generated more quickly than the ZLQI field. According to further aspects of the embodiments, more range (weight) is put on fields whose problems have historically generated more usability complaints. Other factors are maintained (albeit with lesser weight) because they can be indicative of marginal problems that would not always generate a complaint, but which can affect user experience negatively. Installers need the ability to see the effect of changes they are making in their system configuration. In addition, if there are intermittent problems, the user needs to be able to see that the problem is present, or not present, with some

reasonable responsiveness. The use of some or all of the diagnostic data may eventually become part of user interfaces such as Pyng according to further aspects of the embodiments.

According to further aspects of the embodiments, each network device of network **200** can report the ALQI for communications to and from that node; this is accomplished by summing the values of the last eight data fields (which have been numbered as fields **16-23**), and subtracting the same from 255, as shown in Equation 1 below: a high value ALQI indicates a high quality communication link to and from that particular network device.

$$ALQI=255-\min[\Sigma ALQIP,255] \quad (1).$$

According to aspects of the embodiments, the ALQI value is derived by summing the values of the designated ALQIP fields (fields **16-23**), and subtracting the minimum of that summation and 255 from 255. A high value of ALQI means a better communication link in regard to that network device. When summed, the worst case values of the ALQIP fields can total more than 255 (their sum can be as high as 288): the ALQIP fields, described in greater detail below, are APF (with a value range from {0-64}), CCA (with a value range from {0-48}), APR (with a value range from {0-64}), MCF (with a value range from {0-32}), MCR (with a value range from {0-32}), RXC (with a value range from {0-16}), NTT (with a value range from {0-16}), and MTT (with a value range from {0-16}). Allowing the penalty sum to exceed 255 is done primarily to simplify the calculations involved when emphasizing certain fields over others. That is, if all field values were given equal weight, and could not sum greater than 255, the integer division needed to accurately produce fractional scaling becomes "messy." At small ranges, division truncation would also cause small penalty values to get lost. Still further according to aspects of the embodiments, creating the calculation as shown and embodied in Equation 1 ensures that the granularity at small penalty values is maintained while still allowing larger spans to be available to illustrate bad problems. In the unlikely event that the sum of the values exceeds 255, a zero ALQI is reported.

As described in detail above, fields **16-23** are the fields that are used to generate the ALQI value that is reported in field **3**. Fields **16-23** are similar to, or based on, fields **10**, **13**, **11**, **8**, **9**, **12**, **14**, and **6**, respectively (16≈10; 17≈13; 18≈11; 19≈8; 20≈9; 21≈12; 22≈14; 23≈6) albeit with certain different definitions as to each of their derived values, e.g., cumulative versus shorter time intervals. According to further aspects of the embodiments, while fields **16-22** are or can be calculated or determined over one minute periods, if the nature of the node is such that it is asleep, it will not awake just to report an ALQI value; instead, it will awaken, and then after about the first minute, and every minute while awake, will calculate its respective value.

Field **16** is defined as the APF field, and is based on field **10**, the APS failure rate (APSFR). As described above, APSFR is the percentage of APS messages (application messages) that fail to be delivered to gateway device **202**. The APSFR provides a level of confidence that a message will be delivered. In field **16**, however, which comprises an 8 bit signed data field, the APF value is calculated about every minute based on the last minute of APF failures, according to an aspect of the embodiments, instead over a total runtime. The difference is that in the former, the APF value exhibits greater volatility as it will change faster; over time, the APSFR value is time averaged over longer and longer periods of time, and so fluctuates much slower. APF is assigned a range of values from {0-64}.

Field **17** is defined as the CCA field, and is based on field **13**, the BUSY field. As described above, BUSY is the approximate percentage of time that the airspace around a network device is considered "busy" and therefore the device cannot talk. For the CCA field, the CSMA block history average is saturated (i.e., limited to a maximum value of) every minute to a 40% maximum, then scaled to a maximum of 48. Carrier sense multiple access (CSMA) is a probabilistic media access control (MAC) protocol in which a node verifies the absence of the traffic before transmitting on a shared transmission medium, such as an electrical bus, or a band of the electromagnetic spectrum. Carrier sense means that a transmitter uses feedback from a receiver to determine whether another transmission is in progress before initiating a transmission. That is, it tries to detect the presence of a carrier wave from another station before attempting to transmit. If a carrier is sensed, the station waits for the transmission in progress to finish before initiating its own transmission. In other words, CSMA is based on the principle "sense before transmit" or "listen before talk". According to aspects of the embodiments, CCA is calculated over a one minute runtime, and is assigned a range of values from {0-48}, and is based on the last minute of the BUSY rate. According to further aspects of the embodiments, IEEE 802.15.4 PHY provides the following capability to perform CCA according to at least one of the following three methods. CCA Mode 1 defines an energy level above a predetermined threshold. The CCA can report a busy medium upon detecting any energy above the ED threshold. In CCA Mode 2, there is a carrier sense only. The CCA can report a busy medium only upon the detection of a signal with the modulation and spreading characteristics of IEEE 802.15.4. This signal can be above or below the ED threshold. In CCA Mode 3, which is a carrier sense with energy above threshold, the CCA can report a busy medium only upon the detection of a signal with the ED threshold.

Use of field **17** in determining ALQI has advantages according to aspects of the embodiments. If it is seen that node "A" has a high "BUSY" rate, it is having trouble transmitting because of energy/traffic in its vicinity. If device "B" is using "A" in its path, and "B" has a high APS (end-to-end) transmit failure rate, it could be the result of the conditions at device "A" (i.e. "A" is unable to pass messages along due to its local energy problems). This can be a significant troubleshooting aid, because normally all that can be ascertained is that device "B" has messages failing, and so it would otherwise appear the problem is at "B".

Field **18** is the APR field, and is based on field **11**, which reports the APS layer retry rate (APSRR). The APSRR is the percentage of APS send events that are retries for an existing message. In field **18** the APR value is calculated over a one minute block according to an aspect of the embodiments, instead of total runtime, and scaled to a value of {0-64}. The APR value is based on the last minute of APS retries.

Field **19** is the MCF field, and is based on field **8**, which is the MAC failure rate (MACFR) field. The MACFR field indicates the percentage of MAC transmission attempts that fail. In field **19**, the value is calculated over a one minute block instead of over the total runtime according to an aspect of the embodiments. MCF is assigned a value between {0-32}. According to a further aspect of the embodiments, MCF is based on the last minute of MAC failures.

Field **20** is the MCR field, and is based on field **9**, which is the MAC retry rate field (MACRR). MACRR indicates the percentage of MAC transmission attempts that are

retries. The MCR field in field **19** calculates the MAC retry rate over a one minute block instead of over the total runtime, according to an aspect of the embodiments. The MCR field is assigned a value between 0 and 32. According to a further aspect of the embodiments, MCR is based on the last minute of MAC retries.

Field **21** is the RXC field, and is based on field **12**, which is the RXCORR field. RXCORR is based on counts of messages that were received by the network device, but are unable to be decoded properly because of bit errors. In field **21**, RXC is calculated over a one minute block instead of the total runtime, according to an aspect of the embodiments.

The RXC field is assigned a value between 0 and 16. According to a further aspect of the embodiments, RXCF is based on the last minute of RX corruption events.

Field **22** is the NTT field, and is based on field **14**, which is the network turnover (NTRNVR) field. NTRNVR reports the amount of turnover in the network device's neighbor table. In field **22**, the NTT value is calculated over one minute blocks as opposed to the total runtime, according to an aspect of the embodiments. NTT is assigned a value between 0 and 16. According to a further aspect of the embodiments, NTT is based on the last minute of neighbor turnover events.

Field **23** is the MTT field, and is based on field **6**, which is the TBMT field. TBMT is the average time between map-to-one route requests received (MTORR RXs) in seconds. That is, the TBMT field reports the average number of seconds between receiving MTORR messages. MTT is assigned a value between {0-16}. Under normal conditions, TBMT is assigned a maximum value of about {180}. Therefore, the MTT ALQI penalty expression is defined to be:

$$MTT_{ALQIP}=(180-\text{TBMT})\times 3/32 \quad (2)$$

According to Equation (2), the MTT ALQI penalty gets bigger as the TBMT value gets smaller, and scaled to 3/32 for a max of 16 (which is a form of truncated division). In other words, a TMBT of 180 produces an MTT penalty of 0, because this is the network condition that is desired according to aspects of the embodiments. MTT is returned as having a value of zero if no problems occur. Field **23** is slightly different from fields **16-22** in its determination in that it is based not on a strict or somewhat strict time sense, but is instead based on a number of events. That is, field **23**, which reports the MTORR penalty, is based on the last four MTORR events. If less than four events have been reported, then the field's entry with respect to the ALQI summation is zero. If an unusually long period of time transpires since the last fourth event, then the input from this field again is simply maintained on the last four events. If more than four events have occurred since the last reset, the most recent four are used to calculate

Fields **16-23** are used to report the distribution of ALQI penalties for a network device, or node. As described above, the ALQI system, according to aspects of the embodiments, assigns a range and weighting criteria to each of the indicated problems (i.e., the subject matter of each of the fields). The ALQI starts at 255 (FF(H)) for no issues, and penalties are assessed based on the magnitude of each issue indicated. The penalties are summed and subtracted from 255. According to aspects of the embodiments, fields **16-23** as defined above provide an easier indication of communication problems within network **100**.

According to aspects of the embodiments, each of fields **16-23** have been assigned ranges of values of {0-16}, {0-48}, or {0-64}. In all of these cases, the higher the value, the worse the communication issue addressed by the respective field. That is, for example, a value of {64} in field **16**, the APF field (APS transmitter failure), means that a very high percentage of APS messages (application messages) fail to be delivered to gateway device **202**. As described in regard to Equation (1), the sum of the values of field **16-23** comprise the value of ALQIP; to determine the value of ALQI, the minimum of 255 or ALQIP is subtracted from 255; values of ALQI that approach zero indicate communication link problems between and among network devices of PAN **200**. A relatively higher ALQI value, one that approaches 255, indicates a better state of communication links between and among network devices of PAN **200**. According to aspects of the embodiments, a value of ALQI of about 240 and above indicates a robust PAN **200** in terms of communications between and among network devices of PAN **200**; a value of ALQI of about 200 and below indicates a problem in terms of communications between and among network nodes of PAN **200**. However, as those of skill in the art can appreciate, these ALQI numbers are examples only, provided for the purposes of illustration only, and should not be taken in a limiting manner. According to further aspects of the embodiments, different values of ALQI can be ascertained to represent good and bad communications between and among network devices of PAN **200**, based on different types of devices, their robustness in terms of communications, and other factors.

According to further aspects of the embodiments, two or more of fields **16-23** are interrelated; that is, a poor performance in one field-factor will show up as poor performance in a related field. This means that groups of communication issues are related to each other; if a first member of the groups indicates a good communication interface (by a better number (i.e., closer to zero)), then the related member will also indicates a good communication interface. Similarly, if a first member of the group indicate a poor communication interface, then a second member of the group could also indicate a poor communication interface.

According to further aspects of the embodiments, the distinction between good communication values and poor communication values is not necessarily linear; that is, for example, a range of values that indicates good communication values for a certain field can range from {0-5} while the range of poor communication values for the certain field can range from {6-16}.

As an example of what could be done with such data, if we see node "A" has a high "busy" rate, it is having trouble transmitting because of energy/traffic in its vicinity. If device "B" is using "A" in its path, and "B" has a high APS (end-to-end) transmit failure rate, it could be the result of the conditions at device "A" (i.e. "A" is unable to pass messages along due to its local energy problems). This can be a significant troubleshooting aid, because normally all we know is device "B" has messages failing, and so it would otherwise appear the problem is at "B".

FIG. **8** illustrates a flowchart of method **800** for diagnosing communication links in personal area network according to aspects of the embodiments. Method **800** begins with method step **802**, in which a network node of PAN **200**, such as, but not limited to, one or more of network devices **204**, **206**, **208**, **210**, **251**, **252**, **254**, **256**, **258**, **260**, **262**, **264**, and **266**, substantially continuously monitors a first subset of communications link data between itself and other network nodes, as well as between itself and gateway device **202**, based on a first time period. According to aspects of the embodiments, the first time period can be about a minute. According to further aspects of the embodiments, the first

time period can be other time limits, such as less than a minute, several minutes, among other time periods. As described above, the first subset of communications link data can include communication link data such as that shown and described in fields **16-22** of FIGS. **7A-7C**.

In method step **804**, according to a first predefined time interval, one or more of the network nodes generates an ALQI value according to Equation (1) as described in greater detail above. According to aspects of the embodiments, the first predefined time interval can be about a minute (or about 60 seconds), but can be different time intervals as well, and can further be variable between two or more time intervals. The network node generated ALQI value is reported, along with the values of the variables that make up ALQIP as defined herein, as well as the values for the variables that make up ZLQI (and the most recent ZLQI value) are reported to gateway device **202**. According to aspects of the embodiments, ZLQI is constantly updated, but on a time-average basis for the entire run time of each network node. As those of skill in the art can appreciate, ZLQI is updated by the stack layer on any incoming message; ZLQI is maintained for each device (node) in PAN **200** in the neighbor table, but is only reported for the node being used in the routing.

In method step **806**, gateway device **202** receives the ALQI and evaluates it, and takes corrective actions if necessary. According to aspects of the embodiments, gateway device **202** can change channels, and/or provide suggestions in regard to corrective actions (e.g., increasing transmission power, receiver sensitivity, among other types). As those of skill in the art can appreciate, the nodes of PAN **200** can all be operating perfectly correctly and normally, and yet PAN **200** can still be experiencing communications difficulties due to external influences, such as electrical interference caused by machinery, or other sources (i.e., environmental sources).

In method step **808** one or more of the network nodes substantially continuously monitors a second subset of communications link data between itself and other network nodes, as well as between itself and gateway device **202**. In method step **810**, one or more of the network nodes generates a modified ALQI (second ALQI) when a predetermined number of occurrences of the second subset of communications link data occurs. According to an aspect of the embodiments, such predetermined number of occurrences can be four occurrences of the data embodied as field **23** of FIG. **23**. That is, according to an aspect of the embodiments, the second subset of communications link data includes that of field **23** of the data packet reported to gateway device **202** by each of the one or more network nodes of PAN **200**.

In method step **812**, gateway device **202** receives the modified ALQI and evaluates it, and takes corrective actions if necessary. According to aspects of the embodiments, gateway device **202** can change channels, and/or provide suggestions in regard to corrective actions (e.g., increasing transmission power, receiver sensitivity, among other types). As those of skill in the art can appreciate, the nodes of PAN **200** can all be operating perfectly correctly and normally, and yet PAN **200** can still be experiencing communications difficulties due to external influences, such as electrical interference caused by machinery, or other sources (i.e., environmental sources).

In method step **814**, according to a second predefined time interval, each of the one or more network nodes evaluates ZLQI as defined above, and reports the same to gateway device **202**. As described above, the ZLQI values are based on a substantially continuous time averaged value of the variable over the entire running period of each network node (device). According to further aspects of the embodiments, the reporting of ZLQI can be done without consideration to any predefined time period, e.g., the values are updated at the network node whenever new variables values are received (represented by fields **5-15**) and a report is made to gateway device **202**. In method step **816**, gateway device **202** stores and evaluates the reported ZLQI and takes corrective action if possible/necessary.

INDUSTRIAL APPLICABILITY

To solve the aforementioned problems, aspects of the embodiments are directed towards a unique system in which a wireless network reports a series of metrics to be employed for diagnosing communications links between the devices of the wireless networks, such as PAN **200**.

The invention claimed is:

1. A method for determining an operational status of a wireless network system, comprising:

monitoring a plurality of quality of service (QoS) communication indicators by a node of at least two or more nodes of the network system;

generating a first application link quality indicator (ALQI) by the node based on a first subset of the plurality of QoS communication indicators according to a first time period;

modifying the first ALQI with a second subset of the plurality of the plurality of QoS communication indicators only when a first predetermined threshold number of occurrences of the second subset of the plurality of QoS communication indicators has occurred without regard to the first time period to form a first modified ALQI; and

performing a corrective action in regard to the wireless network system that includes at least one of changing channels, increasing transmission power, and increasing receiver sensitivity.

2. The method according to claim **1**, further comprising:

transmitting the modified ALQI by the first node to a second node in the network system.

3. The method according to claim **2**, wherein

the first node is any one of a plurality of network nodes that can include a fully functioning node, and a reduced functioning node, and further wherein,

the second node is a gateway node, and further wherein the method further comprises:

determining a corrective action by the gateway device based on the modified ALQI.

4. The method according to claim **3**, wherein

the corrective action can include one or causing the first node to change channels, increase transmission power, and increase receiver sensitivity.

5. The method according to claim **3**, wherein

the gateway node can communicate one or both of the received ALQI and modified ALQI to a diagnostic console connected via a wireless or wired connection to the gateway node.

6. The method according to claim **1**, wherein the step of generating the first ALQI comprises:

evaluating the following expression—

$$ALQI=255-\min[\Sigma ALQIP, 255],$$

wherein ALQIP comprises the first subset of the plurality of QoS communication indicators.

7. The method according to claim **1**, wherein the step of generating the modified ALQI comprises:

evaluating the following expression—

$$ALQI=255-\min[\Sigma ALQIP, 255],$$

wherein ALQIP comprises the first and second subset of the plurality of QoS communication indicators.

8. The method according to claim **7**, wherein

the number of occurrences of the second subset of the plurality of QoS communication indicators is four.

9. The method according to claim **8**, wherein

the first subset of QoS communication indicators comprises fields **16-22** of a link debug data reporting packet that is generated and transmitted by the first node to the second node, and

the second subset of QoS communication indicators comprise field **23** of the link debug data reporting packet.

10. The method according to claim **9**, wherein

field **16** of the link debug data reporting packet is defined as an APF field, and is based on an APS failure rate (APSFR), and is assigned a range of values from 0 to 64, and wherein APSFR is defined as the percent of application messages that fail to be delivered from the first node to the second node;

field **17** of the link debug data reporting packet is defined as an CCA field, and is based on a BUSY rate, and is assigned a range of values from 0 to 48, and wherein the BUSY rate is an approximate percentage of time that airspace around the first node is considered "busy" and therefore unable to communicate;

field **18** of the link debug data reporting packet is defined as an APR field, and is based on an APS layer retry rate (APSRR), and is assigned a range of values from 0-64, and wherein APSRR is the percentage of APS send events that are retries for an existing message;

field **19** of the link debug data reporting packet is defined as an MCF field, and is based on a MAC failure rate (MACFR), and is assigned a range of values from 0 to 32, and wherein MACFR is the percentage of MAC transmission attempts that fail;

field **20** of the link debug data reporting packet is defined as an MCR field, and is based on a MAC retry rate (MACRR), and is assigned a range of values from 0 to 32, and wherein MACRR is the percent of MAC transmission attempts that are retries;

field **21** of the link debug data reporting packet is defined as an RXC field, and is based on an amount of corrupt received messages (RXCORR), and is assigned a range of values from 0 to 16;

field **22** of the link debug data reporting packet is defined as an NTT field, and is based on a network turnover (NTRNVR), and is assigned a range of values from 0 to 16, and wherein NTRNVR is the amount of turnover in the first node's neighbor table; and

field **23** of the link debug data reporting packet is defined as an MTT field, and is based on time between map-to-one route requests (TBMT), and is assigned a range of values from 0 to 16, and wherein the TBMT is an average number of seconds between receiving map-to-one requests received.

11. The method according to claim **10**, wherein

the MTT can be calculated according to the following expression—

$$MTT_{ALQIP}=(180-TBMT)\times 3/32$$

wherein,

MTT is not reported unless a number of occurrences of TBMT is greater than or equal to four, and further wherein

MTT is determined based on the last four occurrences of TBMT, regardless of the number of TBMT occurrences.

12. The method according to claim **1**, wherein

the first time period is about one minute.

13. The method according to claim **2**, further comprising:

generating a zigbee link quality indicator (ZLQI) based on a third subset of the plurality of the plurality of QoS communication indicators.

14. The method according to claim **13**, wherein

the third subset of QoS communication indicators comprises fields **5-15** of a link debug data reporting packet that is generated and transmitted by the first node to the second node, and wherein

ZLQI is maintained over a runtime of the first node.

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